

# GLOBAL MACRO INTELLIGENCE BRIEF

2026 Edition

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Twenty-one under-covered risks to the global financial system, regrouped by mechanism into five sections.

*Compiled 29 June 2026 · Informational analysis — not investment, legal or financial advice.*

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## How to read this brief

**What this is.** Twenty-one under-covered risks to the global financial and economic system, each written to be readable by a non-specialist and useful to a professional. They were chosen because they are consequential, connected, and under-discussed relative to their importance.

**How it is organized.** The stories are grouped into five mechanism-based sections — Funding & Leverage; Credit, Consumers & Valuations; Sovereigns & the Monetary Order; Geoeconomics & Trade Weapons; and Physical & Resource Limits. Grouping by mechanism, rather than by headline, is what makes the connections between stories visible, and it is the same structure the live tracker uses to score risk.

**How to read each story.** Every story opens with who it affects and a threat meter (1–10, from Low to Severe), then explains what is happening, why it is under-covered, the genuine debate on both sides, a stress test, a historical parallel, a plain-English summary, and what an ordinary person can reasonably do.

**A note on the levels.** The 1–10 score measures vulnerability — the potential to do systemic damage if stressed — not the probability of a near-term blow-up. A high number is a reason to watch, not to panic.

### Confidence tags

**[Established]** widely documented by multiple authoritative sources.

**[Contested]** real, but debated in size or interpretation.

**[Single-source]** reported but not yet widely corroborated.

**[Forecast]** a forward projection, not a present fact.

## Risk dashboard — 21 stories at a glance

Each story's current threat level on the 1–10 scale (Low · Moderate · Elevated · High · Severe). Levels measure vulnerability if stressed, not probability. Section averages are shown in brackets.

### Funding & Leverage — avg 4.8/10

- 1. The Treasury “basis trade”: hidden leverage under the world’s safest asset — 5/10 (Elevated)
- 2. Japan’s \$2.3tn bet, the JGB market, and the carry-trade fuse — 6/10 (Elevated)
- 3. Private credit’s quiet \$2tn expansion — and its first cracks — 5/10 (Elevated)
- 4. Stablecoins quietly become a pillar — and a fault line — of Treasury demand — 4/10 (Moderate)
- 5. Clearinghouses: the financial system’s hidden single point of failure — 4/10 (Moderate)

### Credit, Consumers & Valuations — avg 5.2/10

- 6. The AI build-out’s circular money machine — 6/10 (Elevated)
- 7. The commercial real-estate “maturity wall” and your regional bank — 5/10 (Elevated)
- 8. The hidden consumer-debt squeeze: subprime cars and “phantom” instalments — 5/10 (Elevated)
- 9. The quiet retreat of insurance — and why it threatens your home’s value — 5/10 (Elevated)

### Sovereigns & the Monetary Order — avg 4.0/10

- 10. The silent debt crisis in the developing world — 5/10 (Elevated)
- 11. France: the eurozone’s quiet structural fault line — 4/10 (Moderate)
- 12. Central banks are quietly buying gold and trimming the dollar — 3/10 (Moderate)
- 13. The retirement math nobody wants to say out loud — 4/10 (Moderate)

### Geoeconomics & Trade Weapons — avg 6.0/10

- 14. China is exporting its deflation — the global transmission nobody frames clearly — 6/10 (Elevated)
- 15. Critical minerals as a weapon: “availability is not access” — 6/10 (Elevated)
- 16. The lithography chokepoint: one machine the whole digital world needs — 5/10 (Elevated)
- 17. Sea lanes and undersea cables: the physical arteries of globalization — 7/10 (High)

### Physical & Resource Limits — avg 5.0/10

- 18. AI’s power problem: the grid is the new bottleneck — 6/10 (Elevated)
- 19. Copper: the metal the electric future can’t do without — 4/10 (Moderate)
- 20. Water: the economy’s most underpriced input — 4/10 (Moderate)
- 21. Food and fertilizer: a concentrated, fragile system — 6/10 (Elevated)

## Section 1 — Funding & Leverage

These five stories share a single mechanism: leverage that lives in the plumbing of the financial system, funded by someone's very short-term money. The Treasury basis trade, Japan's carry trade, private credit, stablecoins, and the clearinghouses that sit behind them are rarely discussed together, but they fail the same way — through funding that can evaporate faster than the assets it supports can be sold.

The basis trade and the clearinghouses are mechanically linked: the same Treasury positions that hedge funds finance overnight in repo are the ones moving into central clearing, so a margin spiral in one is a margin spiral in the other. Japan's carry trade and stablecoins are the cross-border and crypto-native versions of the same bet — borrow cheap and short, hold something longer, and hope the funding rolls. Private credit is the slow-motion cousin: less liquid and less marked-to-market, but levered against the same business cashflows. What ties the set together is reflexivity — a shock to funding costs forces selling that worsens the very prices that triggered it. This is the part of the system that broke in March 2020, and that the post-crisis clearing reforms are racing to contain before it breaks again.

## 1. The Treasury “basis trade”: hidden leverage under the world’s safest asset

**WHO THIS HITS:** *Anyone with savings, a mortgage, or a pension — this sits under the US Treasury market, which sets the price of almost all borrowing and most investments.*

**CURRENT THREAT LEVEL:** **5/10 — Elevated** *Systemic but currently orderly*

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**Why this level:** Among the largest hidden-leverage points in markets, but it has stayed orderly through recent volatility and central clearing is on the way. **Raises it:** A spike in Treasury volatility (MOVE) or repo rates forcing margin calls; a delay or carve-out to SEC central clearing. **Lowers it:** Central clearing implemented on schedule; falling Treasury volatility; a shrinking share of zero-haircut repo.

### What is happening

**[Established]** A small group of highly leveraged hedge funds has become a central marginal player in the \$30tn US Treasury market through the cash-futures basis trade — buying cash Treasuries, shorting Treasury futures, and financing the position in the overnight repo market. The Dallas Fed reports hedge-fund net repo borrowing reached roughly \$1.8tn by year-end 2025, about 6% of marketable notes and bonds.

**[Contested]** Estimates of the trade’s size vary by method; some gauges based on gross futures exposure put it near \$1.4tn, potentially around twice its 2020 scale. **[Established]** What is not disputed: leverage is concentrated and high. The largest Treasury-holding funds run balance-sheet leverage well above the sector average, and a Federal Reserve analysis found the great majority of hedge-fund repo borrowing transacts at zero or negative haircuts — meaning almost no cushion if prices move.

**[Established]** Fed Governor Lisa Cook publicly flagged these positions as a systemic vulnerability in November 2025. The SEC’s mandate to centrally clear cash Treasuries (end-2026) and Treasury repo (mid-2027) is the regulators’ main structural response — still ahead, not yet in force.

### Why it is under-covered

It is pure market plumbing with no retail hook, no villain, and a name that signals “technical.” Yet it sits beneath the asset that prices everything else on Earth. When it breaks, it does so violently and the public only learns the term afterwards.

### The debate

**Regulators argue:** the trade is large, concentrated, funded by ultra-short repo, and exposed to a doom loop — a volatility spike raises futures margins and repo haircuts at once, forcing funds to dump cash Treasuries into a falling market, widening the very basis they were arbitraging.

**The industry argues:** the trade is functionally useful — it keeps futures and cash prices aligned, adds auction demand and liquidity in normal times — and that fund-level leverage is far below 2008-era securitisation chains.

**Both sides accept:** the position is genuinely large, genuinely concentrated in a handful of funds, central to Treasury functioning, and that central clearing is coming. The disagreement is only about how it behaves under acute stress.

### Stress test (the benign case)

The optimistic read held up in April 2025: NY Fed officials noted the basis trade stayed orderly through tariff-driven volatility. Central clearing, once live, should mutualise risk and reduce zero-haircut

financing. If the Fed's Standing Repo Facility back-stops funding on demand, a 2020-style seizure may simply not recur. The weakness in that case: the facility is untested at scale during a true panic, clearing is still 12–18 months away, and concentration means a few funds failing simultaneously could still overwhelm dealer balance sheets.

### Historical parallel

March 2020: basis-trade unwinding was a major contributor to the Treasury market dysfunction that forced the Fed into massive emergency purchases. Further back, the 1998 collapse of Long-Term Capital Management showed how a relative-value arbitrage on tiny spreads, levered enormously, can threaten the system when the spread moves the “wrong” way and convergence arrives too late to meet today's margin call. Both episodes ended with official intervention, not market self-correction.

### In plain terms

A handful of hedge funds have borrowed enormous sums to run a razor-thin-margin trade in US government bonds — the safest, most important market in the world. In calm times it helps the market work. But it is so leveraged that a sudden jump in volatility could force them to dump bonds all at once — which is what briefly broke this market in March 2020 and required an emergency Fed rescue. It isn't flashing red now, but it's a hidden fragility sitting under everything else.

### What you can do now

**Keep an emergency cash cushion.** If the bond market seizes up, everything wobbles for a while; cash means you aren't forced to sell at the worst moment.

**Don't assume “bonds are always safe.”** In a 2020-style episode even US Treasuries briefly fell; spreading across asset types matters more than owning any one ‘safe’ thing.

**Favor simple, low-cost, broad funds** over leveraged or exotic bond products you don't fully understand.

**Mostly, this is awareness.** It's a plumbing risk you can't personally fix — but knowing why a sudden bond-market air-pocket can happen helps you avoid panic-selling into one.

### What to watch

- The SOFR-fed funds spread and repo rates spiking at quarter-ends: signals dealer balance-sheet strain and tightening funding for the trade.
- The MOVE index (Treasury volatility): sustained spikes are what trigger margin/haircut doom loops.
- SEC Treasury-clearing implementation milestones through 2026–2027: delays or carve-outs would leave the vulnerability intact.
- Form PF and OFR repo data showing whether leverage and zero-haircut share keep rising.





## 2. Japan's \$2.3tn bet, the JGB market, and the carry-trade fuse

**WHO THIS HITS:** Anyone invested in global stocks or bonds — a sudden unwind of Japan's 'carry trade' can jolt markets worldwide, as it briefly did in August 2024.

CURRENT THREAT LEVEL: **6/10 — Elevated** *Live, time-sensitive repricing*

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**Why this level:** JGB yields at multi-decade highs, a huge new spending plan with unclear financing, and a recent (August 2024) carry-unwind precedent make this the most time-sensitive story in Set 1. **Raises it:** A sharp yen surge alongside rising JGB yields; messy financing of the Takaichi plan; a hawkish BoJ surprise. **Lowens it:** Orderly yield rises with a stable yen; a credible funding path for the plan; calm global carry conditions.

### What is happening

**[Established]** On 24 June 2026, Prime Minister Sanae Takaichi unveiled a long-term plan to mobilise more than ¥370tn (about \$2.3tn) of public and private investment over 14 years to March 2041, with ¥101.6tn earmarked for AI and semiconductors. Government is expected to fund a little under half. The plan landed with Japanese equities euphoric — the Nikkei 225 briefly topped 70,000 for the first time this month — and bond strategists uneasy.

**[Established]** The bond market context is the real story. The Bank of Japan has been normalising after decades of ultra-loose policy: it ended yield-curve control and negative rates, is running quantitative tightening, and holds a 0.5% policy rate. Long-end yields have repriced sharply — the 30-year JGB hit a record 3.875% in January 2026 — and life insurers, the traditional buyers of super-long bonds, have been trimming exposure, prompting Japan's FSA to accelerate a review of unrealised losses on their balance sheets.

**[Single-source]** One market write-up puts Japan's debt-to-GDP near 232% for end-2026; **[Established]** the government's own three fiscal scenarios show the debt ratio declining only in the optimistic case and rising again in the 2030s otherwise. Bloomberg's same-week assessment was blunt: the plan "raises fresh JGB concerns."

### Why it is under-covered (in the West)

It reads as a domestic Japanese fiscal story. But Japan is the world's largest creditor nation, and the yen is the world's funding currency. The transmission belt from Tokyo's bond market to global asset prices runs through the carry trade — and that belt is rarely on Western front pages until it snaps.

### The debate

**The bull case:** decisive industrial policy (AI, chips, defence, shipbuilding) finally breaks Japan out of stagnation; reflation plus growth shrinks the debt ratio; equity markets are voting yes.

**The bond-market case:** the plan's financing is unspecified, issuance pressure meets a BoJ that is buying less, and rising domestic yields plus a stronger yen make Japanese institutions repatriate capital.

**Both sides accept:** the plan is unprecedented in scale, the financing path is genuinely unclear, and long-end JGB yields are structurally higher than at any point in a generation.

### Stress test (the benign case)

Japan funds itself almost entirely domestically, in its own currency, with a vast pool of household and institutional savings — the textbook reason a 230%-plus debt ratio has never triggered a crisis. If

reflation is real and wages keep rising, higher nominal growth makes the debt easier, not harder, to carry, and insurers eventually become natural buyers of higher-yielding super-longs again. The vulnerability in that case is timing and speed: an orderly repricing is benign; a disorderly one that coincides with a yen surge forces a synchronised global carry unwind.

### Historical parallel

August 2024 is the live warning: a modest BoJ hike plus yen strength triggered a violent, if brief, global carry-trade unwind that hit equities worldwide. Late 2022–early 2023 saw global bond markets wobble the last time JGB yields jumped meaningfully. The pattern is consistent — yen-funded leverage built quietly over years can reverse in days.

### In plain terms

For decades Japan had near-zero interest rates, so investors borrowed cheaply in yen to buy higher-returning assets all over the world — the ‘carry trade.’ Now Japan’s rates and bond yields are rising and it has just announced a giant spending plan. If the yen jumps and Japanese money heads home, that global trade can unwind fast, yanking money out of markets everywhere in a matter of days. A mini-version of exactly this happened in August 2024.

### What you can do now

**Expect occasional sharp, short drops** from this kind of unwind — and don’t panic-sell into them; historically they’ve been violent but brief.

**Keep a cash buffer** so a sudden global wobble doesn’t force your hand at the bottom.

**Diversify globally** rather than concentrating in the assets most reliant on cheap-yen funding (high-flying tech, some emerging markets).

**If you hold Japan or currency-hedged funds**, understand that a stronger yen changes how they behave.

### What to watch

- USD/JPY together with the 10- and 30-year JGB yields: rapid yen strengthening plus rising yields is the classic carry-unwind signature.
- Japanese investors’ net flows in foreign bonds (especially US Treasuries): repatriation removes a major marginal buyer of US debt — directly linking this story to Story 1.
- BoJ meeting guidance and the pace of QT; any sign the Ministry of Finance must lean on issuance to fund the plan.
- Life-insurer disclosures and FSA findings on unrealised bond losses.



### 3. Private credit's quiet \$2tn expansion — and its first cracks

**WHO THIS HITS:** Anyone with money in pensions, insurance products, or 'alternative' funds — private credit has quietly spread into retirement and insurance portfolios.

**CURRENT THREAT LEVEL:** 5/10 — **Elevated** *First cracks, slow shakeout*

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**Why this level:** Defaults are rising from a low base and high-profile blow-ups have appeared, but lock-ups and lower leverage point to a grind rather than a sudden break. **Raises it:** A correlated AI/software default wave; redemption gates at large funds; a widening gap between private marks and public prices. **Lowens it:** Defaults plateauing near historical norms; transparent, timely markdowns; no insurer-channel contagion.

#### What is happening

**[Established]** Private credit — non-bank direct lending to mid-sized companies — has grown to roughly \$1.5–\$2.5tn in assets and now sits largely outside the prudential regulatory perimeter. In May 2026 the Financial Stability Board warned that its complexity, leverage, valuation opacity and interconnection with banks and insurers could amplify stress. The European Central Bank and Bank of England have echoed the concern; the BoE is stress-testing the sector.

**[Established]** The first real cracks appeared in 2025 with the collapses of First Brands and Tricolor in US auto-parts and asset-based finance, which pushed the question of hidden risk in private markets into view. Morgan Stanley warned in early 2026 that direct-lending default rates could climb toward 8% — versus a 2–2.5% historical average — with pressure concentrated in software borrowers exposed to AI disruption, while adding the spike would be “significant but not systemic.” Use of payment-in-kind arrangements, where borrowers defer cash interest, has risen — a classic stress release valve that also masks strain.

**[Established]** The most under-watched channel is the private-equity-to-insurance linkage: insurers affiliated with PE firms hold large private-credit concentrations, and a small but growing private-credit CLO market (around 3% of assets) is an early step back toward securitisation. Fed Governor Cook named private credit one of four key financial-stability risks.

#### Why it is under-covered

It is private by design. There are no daily prices, no public ratings on most borrowers, and net asset values are reported with a 60–90 day lag using model-based marks. Opacity is the product feature that makes the risk hard to see — and hard to report.

#### The debate

**The warning case:** lower-quality, higher-leverage borrowers; valuations that lag reality; insurer balance sheets and retirement savings quietly exposed; a “zero-loss” marketing myth meeting its first downturn.

**The reassurance case (bank CEOs, Treasury):** fund-level leverage is modest (roughly 1–1.5x versus 30–40x in pre-2008 securitisation), capital is locked up for years rather than runnable, and the market is “still small enough” to avoid systemic threat.

**Both sides accept:** defaults are rising from low levels, valuation practices are opaque, and the sector is untested through a full, prolonged downturn.

### Stress test (the benign case)

The structural differences from 2008 are real: long lock-ups prevent depositor-style runs, leverage is far lower, and there is no comparable web of CDOs. The most probable adverse path is therefore not a sudden collapse but a slow shakeout — rising defaults, forced markdowns, restricted redemptions, tighter credit for mid-sized firms — painful for investors and borrowers but absorbable by the banking system, which the Fed's 2025 stress test suggested could withstand even severe non-bank stress. The risk to that benign read is correlation: if an AI-driven revenue shock hits software borrowers all at once, diversification fails exactly when it is needed.

### Historical parallel

The unavoidable comparison is subprime — opaque, rapidly grown, model-valued credit — but the analogy is contested and imperfect given the leverage and liquidity differences. A closer fit may be the slow-burn savings-and-loan and leveraged-loan workouts: not a single detonation, but a multi-year grind of markdowns and restructurings that traps capital and tightens lending well after the headlines fade.

#### In plain terms

Big investment firms have built a roughly \$2 trillion business lending directly to mid-sized companies — outside the banks, and outside daily market prices. Because there are no public prices, trouble stays hidden until it surfaces, and the first cracks (a couple of large collapses) have appeared. It's more likely to be a slow grind of losses than a sudden crash — but a lot of it has crept into insurance and retirement money, so ordinary people are exposed without realizing it.

#### What you can do now

**Check your exposure.** Ask whether your retirement or insurance products hold 'private credit' or 'alternatives,' or read the fund factsheet.

**Be skeptical of 'high steady yield, no volatility.'** Smooth returns often just mean the price isn't being updated — not that there's no risk.

**Mind the lock-ups.** Many of these funds can trap your money for years and value their holdings on a lag; don't chase the yield without understanding both.

**Diversify.** Don't let one yield-chasing product become a large share of your savings.

### What to watch

- The gap between private marks and public-market comparables (e.g. liquid leveraged-loan and BDC prices): a widening gap understates realised losses.
- Rising payment-in-kind usage and redemption restrictions ("gates") at large funds: signs strain is being deferred rather than resolved.
- Default rates in software/tech-services direct lending specifically — the AI-exposed segment.
- Disclosures on PE-owned insurers' private-credit concentration: the channel into retirement assets.



## 4. Stablecoins quietly become a pillar — and a fault line — of Treasury demand

**WHO THIS HITS:** Anyone who uses crypto or dollar ‘stablecoins’ — and, indirectly, all savers, because these tokens are becoming a real buyer of US government debt.

**CURRENT THREAT LEVEL:** 4/10 — Moderate *Fast-growing but still small*

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**Why this level:** A new and run-prone marginal buyer of T-bills, but still modest relative to the system and freshly regulated under the GENIUS Act. **Raises it:** A major stablecoin losing its peg under stress; rapid, thinly-backed growth; a run forcing T-bill fire-sales. **Lowers it:** Smooth GENIUS implementation; full, transparent reserve backing; concentration in well-capitalised issuers.

### What is happening

**[Established]** Dollar stablecoins grew to roughly \$280bn outstanding by year-end 2025, from about \$25bn in 2020. The July 2025 GENIUS Act gave them a federal framework requiring 1:1 backing in cash and short-dated Treasuries (93 days or less) plus repo, with monthly disclosure. The US Treasury Secretary has openly framed the law as a tool to cement dollar dominance and “surge” demand for US Treasuries, projecting a multitrillion-dollar market.

**[Established]** That makes stablecoin issuers a fast-growing, price-insensitive buyer of Treasury bills — holdings that already rival some sovereign holders. ECB and BIS research finds a \$3.5bn inflow into dollar stablecoins lowers 3-month bill yields by roughly 2.5–3.5 basis points, an effect that more than doubles when bills are scarce. The ECB explicitly describes a “global safe-asset channel” and frets that Europe must field euro stablecoins or cede ground.

### Why it is under-covered

It is filed under “crypto,” which causes serious macro readers to skip it. But the relevant story is not speculation — it is market structure: a new, lightly understood, run-prone intermediary is becoming a material marginal buyer at the short end of the world’s most important bond market.

### The debate

**Boosters argue:** stablecoins extend dollar reach globally, deepen Treasury demand, and modernise payments (instant, programmable, 24/7).

**Skeptics argue:** they import run risk into the Treasury market, lack 24/7 redemption infrastructure, raise illicit-finance questions, and the fiscal benefit is partly offset by lost seigniorage as currency demand falls.

**Both sides accept:** the sector is growing fast, is now macro-relevant to short-end Treasury demand, and concentrates reserves in a narrow band of assets.

### Stress test (the benign case)

GENIUS’s reserve, disclosure and audit requirements are precisely designed to make these instruments boring and run-resistant; full reserve backing in T-bills is far safer than the assets behind past failures. If the rules hold and a few well-capitalised issuers dominate, stablecoins could be a stabilising, structural source of bill demand. The vulnerability: a loss of peg under stress (USD Coin briefly fell to about \$0.87 in March 2023) could force rapid, synchronised T-bill sales into a stressed market — transmitting a crypto shock into Treasury funding, the very plumbing Stories 1 and 2 also run through.

## Historical parallel

The closest analogue is the 2008 “breaking the buck” of the Reserve Primary money-market fund, which triggered a run across prime money funds and a government backstop. Stablecoins are economically similar — fixed-value claims backed by short paper, vulnerable to redemption runs — but without the decades of money-fund regulation and now wired into round-the-clock crypto markets that never close for a cooling-off period.

### In plain terms

‘Stablecoins’ are crypto tokens meant to always be worth \$1, backed by cash and short-term US government debt. They’ve grown fast enough to matter to the Treasury market. The risk: if a big stablecoin ‘breaks the buck’ in a panic (one briefly did in 2023), it could be forced to dump those Treasuries quickly, pushing a crypto shock into the wider financial plumbing. New US rules (the GENIUS Act) aim to make them safer.

### What you can do now

**Remember they’re not bank deposits.** Stablecoins aren’t FDIC-insured, and the ‘\$1’ peg can break under stress.

**Prefer large, transparent, fully-reserved issuers** that publish audits — and don’t park money you can’t afford to lose in them for a little extra yield.

**Don’t treat the yield as ‘risk-free.’** That yield is payment for real risks.

**For most people this is awareness;** if you don’t use crypto, your exposure is indirect (its effect on the Treasury market).

### What to watch

- Total stablecoin supply and the share held in T-bills vs repo: the size of the new marginal buyer.
- Any peg deviation in major stablecoins under market stress — the run trigger.
- GENIUS implementing rules from Treasury/OCC and the Fed’s stance on payment accounts — the guardrails.
- Concentration: how few issuers dominate, and whether non-financial firms enter via loopholes.





## 5. Clearinghouses: the financial system's hidden single point of failure

**WHO THIS HITS:** Everyone with savings in markets — a handful of clearinghouses sit at the center of the financial system, and one failing would be a system-wide event.

**CURRENT THREAT LEVEL:** **4/10 — Moderate** Concentrating risk, no near-term trigger

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**Why this level:** Clearinghouses concentrate the system's residual risk in a few 'too-important-to-fail' hubs, and the US Treasury-clearing mandate will push even more through them — but failures are historically rare and buffers are large. **Raises it:** A volatility spike forcing huge procyclical margin calls (cf. UK 2022 LDI, nickel 2022); a large clearing-member default. **Lowers it:** Smooth Treasury-clearing implementation with robust capacity; stronger CCP recovery and resolution plans; calmer markets.

### What is happening

**[Established]** After 2008, regulators pushed vast swaths of trading through central counterparties (CCPs) — clearinghouses that stand between buyer and seller to guarantee trades. This genuinely reduced the tangled, bilateral counterparty risk that amplified the crisis. But it also concentrated the system's residual risk into a handful of "too-important-to-fail" hubs — names like CME, ICE, LCH, Eurex, and DTCC/FICC. A Chicago Fed analysis finds the clearing ecosystem is now *more concentrated than it was in 2007*, before the crisis.

**[Established]** The mechanism that makes CCPs safe in normal times can make them dangerous in a panic. CCPs require members to post cash margin that rises automatically with volatility — so their liquidity demands are inherently *procyclical*: exactly when markets are stressed and cash is scarce, the clearinghouse calls for more of it, which can force fire-sales and propagate the stress it was meant to contain. The Fed has flagged this systemic liquidity risk explicitly.

**[Established]** The live development: the SEC's mandate to centrally clear US Treasuries (cash by end-2026, repo by mid-2027) — the regulators' fix for the basis-trade risk in Set 1 — will route an enormous additional volume through CCPs, *increasing* concentration. In other words, the cure for one chokepoint (hidden leverage in Treasuries) deepens another (clearinghouse concentration).

### Why it is under-covered

CCPs are deliberately invisible plumbing — the system works precisely because no one thinks about them. There's no retail hook and the machinery is arcane (default waterfalls, margin models, mutualized loss funds). Only three CCPs have ever failed (in 1974, 1983, and 1987), which breeds a complacency that the consequences of a fourth would shatter.

### The debate

**The reassurance camp argues:** CCPs work — they survived 2008, 2020, and 2022 — with deep, pre-funded default buffers, strict margining, and far more transparency than the bilateral web they replaced; concentration brings standardization and oversight.

**The concern camp argues:** concentration plus procyclical margin calls turns CCPs into potential transmission mechanisms; their risk is poorly understood; recovery and resolution plans are untested; and the Treasury mandate is loading still more onto them.

**Both sides accept:** central clearing reduced bilateral counterparty risk but concentrated residual risk, margin demands are procyclical, and the few key CCPs are now critical infrastructure.

### Stress test (the benign case)

The buffers are real and large: pre-funded default funds, layered “waterfalls,” conservative margining, and stress tests designed around multiple simultaneous member defaults. CCPs absorbed the 2020 dash-for-cash and the 2022 commodity and gilt shocks without failing. The vulnerability is the tail no buffer is sized for: a fast, correlated volatility spike that triggers margin calls system-wide (as the UK’s 2022 LDI episode and the 2022 nickel-market chaos foreshadowed), or the failure of a large clearing member — against a backdrop of rising concentration.

### Historical parallel

The clearest warnings are recent and partial: the UK’s September 2022 pension (LDI) crisis, where margin calls on leveraged hedges forced a doom-loop of gilt selling until the Bank of England intervened; and the March 2020 “dash for cash,” when margin demands drained liquidity system-wide. The 1987 crash nearly broke the clearing system and did topple the Hong Kong Futures Exchange’s clearer. The pattern: clearing is invisible and reassuring until a volatility spike turns its safety mechanism into an accelerant.

#### In plain terms

After the 2008 crisis, regulators routed most trading through a few ‘clearinghouses’ — middlemen that sit between buyers and sellers and guarantee trades. This made the system safer in normal times, but it piled the leftover risk into a handful of giant hubs that are now ‘too important to fail.’ The catch: these hubs demand more cash from their members exactly when markets are panicking and cash is scarce — which can make a crisis worse. And a new rule pushing US government-bond trading through them (the fix for a problem in Set 1) is concentrating even more risk in one place. Failures are very rare and the safety buffers are big, so this is a low-probability tail risk — but if it ever broke, it would be a system-wide event.

#### What you can do now

**This is awareness — there’s no direct retail action**, but it’s part of why broad diversification and a cash buffer matter: in a true plumbing crisis, everything sells off together for a while.

**Keep an emergency fund outside the markets.** The recurring lesson of every plumbing scare (2008, 2020, 2022) is that cash on hand means you’re never forced to sell at the bottom.

**Be skeptical of leverage and ‘can’t-lose’ hedges.** The 2022 UK pension blow-up came from leveraged strategies that needed cash exactly when it vanished; the same logic applies to your own finances — avoid being a forced seller.

**Don’t lose sleep over this one specifically.** It’s a low-probability tail risk handled by people whose job is to handle it; the general ‘cash buffer + diversify’ habit is the appropriate response.

### What to watch

- Market volatility (VIX/MOVE) and margin-call episodes — the trigger for procyclical stress.
- US Treasury-clearing implementation (end-2026 cash, mid-2027 repo) and whether clearing capacity scales smoothly.
- CCP stress-test results and recovery/resolution planning from regulators.

- Any large clearing-member distress — the most direct failure path.



## Section 2 — Credit, Consumers & Valuations

If the previous section is about how fast money can leave, this one is about whether the underlying loans and asset values were ever worth what the books say. The AI build-out's circular financing, the commercial-real-estate maturity wall, the hidden consumer-debt squeeze, and the quiet retreat of insurance are all stories of deferred recognition — losses or risks that exist now but have not been booked yet.

The common thread is the gap between mark and reality: AI infrastructure financed against revenue not yet earned; office buildings carried near par that must refinance at double the rate; households whose buy-now-pay-later and subprime balances do not show up in headline credit stats; and homes whose insurability — and therefore value — is silently eroding as insurers withdraw. Each can extend for years on 'extend and pretend.' Each resolves the same way: when refinancing or repricing finally forces the mark to meet reality, the loss arrives all at once. These stories also feed the section before them — a credit shock here is exactly the kind of volatility spike that detonates the leverage there.

## 6. The AI build-out's circular money machine

CURRENT THREAT LEVEL: **6/10 — Elevated** *Large, debt-fuelled, concentrated*

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**Why this level:** Enormous, increasingly debt-financed spending concentrated in a few index-dominating names, with a 2026–27 refinancing wall — though hyperscaler cash flows are strong. **Raises it:** An AI-revenue disappointment; widening bond spreads on the big issuers; GPU-debt refinancing stress. **Lowers it:** AI revenue ramping to meet capex; build-out funded from cash flow; clearer, consistent disclosure.

**WHO THIS HITS:** *Anyone with a pension, 401(k), or index fund — because a handful of AI names now dominate the market you are probably invested in.*

### What is happening

**[Established]** Spending on AI data centres has reached a scale without precedent. Big Tech's 2026 capital expenditure is projected to exceed \$600bn — up roughly 50% after a ~60% jump in 2025 — with several estimates of the broader AI build-out around \$650–700bn for the year. Capital intensity (capex as a share of revenue) has climbed to levels that are historically unheard of for the sector.

**[Established]** Two features make this more than a spending story. First, **circularity**: the same companies invest in, lend to, and buy from one another — chipmakers funding model labs that commit to cloud providers that buy those chips. Supporters call it a virtuous circle that locks in scarce supply; critics call it round-tripping that can inflate apparent demand. Second, **concentration**: the combined weight of Meta, Alphabet, Amazon and Oracle in the US investment-grade corporate bond index nearly doubled over the year to April 2026, from 2.2% to 4.1%, as they issued debt to fund the build-out.

**[Contested]** Estimates of the debt involved vary; one widely cited figure puts buildout-related debt around \$1.5tn over time. **[Forecast]** Consultancy work cited in academic analysis flags a potential ~\$800bn-a-year gap by 2030 between committed infrastructure spending and realised AI revenue. The first wave of debt backed by GPUs (the chips themselves) matures in 2026–2027, just as those chips depreciate.

### Why it is under-covered

The headline — “AI is booming” — is everywhere. The financing plumbing underneath is not: vendor financing, take-or-pay contracts, off-balance-sheet special-purpose vehicles, and supplier financing showing up as rising accounts payable. Disclosure is inconsistent across companies, which is exactly what makes the aggregate risk hard to see.

### The debate

**Bulls argue:** demand for computing is real and exploding; locking in suppliers and customers through paired investment-and-purchase deals is rational in a supply-constrained market (Janus Henderson's “virtuous circle”; KKR's “beyond the bubble”). Hyperscalers are highly profitable and investment-grade.

**Skeptics argue:** circular deals can make demand look bigger than it is; the gap between capex and actual AI revenue is large and widening; GPU-backed debt plus rapid chip depreciation is a refinancing trap; and concentration means a stumble hits the whole index.

**Both sides accept:** the spending is real and enormous, it is increasingly debt-financed, and it is concentrated in a handful of firms whose fortunes now move the broad market.

### Stress test (the benign case)

If AI adoption inflects upward as the bulls expect, today's capex looks visionary rather than reckless: the hyperscalers genuinely can fund much of this from cash flow, their credit is strong, and demand fills the capacity. The weak point in that case is timing and correlation — revenue must ramp before depreciation and debt maturities bite, and because the same few names dominate both the bond and equity indices, a single disappointment transmits everywhere at once. A Moody's scenario of a sharp decline maps contagion into banks via construction loans and GPU-backed facilities.

### Historical parallel

The closest analogue is the 2000–02 telecom build-out, where vendor financing (equipment makers lending customers the money to buy their gear) masked weak end-demand until it collapsed. Academic work also draws on 2008 (counterparty opacity and off-balance-sheet leverage) and the 2014–16 oil bust (physical overbuilding meeting a price collapse). The recurring lesson: infrastructure booms funded by circular, debt-laden financing tend to overshoot, and the bill arrives when financing tightens.

#### In plain terms

The companies building artificial intelligence are spending staggering sums — over \$600 billion this year — much of it borrowed. A worry is that they keep investing in and buying from each other, which can make demand look bigger and healthier than it really is. Because just a few giant companies are doing most of this, they now make up an unusually large slice of the stock and bond markets. If AI doesn't earn money as fast as hoped, the disappointment could hit not just those companies but the index funds most people's retirement savings sit in.

#### What you can do now

**Know what you actually own.** A standard S&P 500 index fund is now heavily weighted toward a few AI-linked giants. "Diversified" may be less diversified than you think — look up your fund's top-10 holdings and their combined weight.

**Diversify deliberately.** Spreading across regions, sectors, and asset types (not just more US tech) reduces how much one AI stumble can hurt you.

**Avoid concentrating in single AI names** or chasing the hype with money you can't afford to lose; resist leverage and "story" bets.

**Keep an emergency cash buffer.** Bubble timing is unknowable; cash on hand means a market drop doesn't force you to sell at the bottom.

### What to watch

- Hyperscaler free cash flow vs capex — the moment build-out is funded by debt rather than profits.
- AI revenue disclosures vs the capex-revenue gap; signs of under-used ("idle") capacity.
- Corporate bond spreads on the big four issuers, and any credit-rating pressure.
- GPU-backed debt maturities in 2026–2027 and refinancing terms.





## 7. The commercial real-estate “maturity wall” and your regional bank

CURRENT THREAT LEVEL: **5/10 — Elevated** *Active distress, widely anticipated*

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**Why this level:** Record office delinquencies and a genuine maturity wall, but the peak is easing and the risk is unusually well-telegraphed. **Raises it:** A cluster of regional-bank losses; office values falling further; ‘extend and pretend’ ending abruptly. **Lowers it:** Orderly refinancing via private credit; office-value stabilisation; the maturity peak passing.

**WHO THIS HITS:** *Anyone who banks with a regional or community bank, works in a downtown economy, or owns property near struggling office districts.*

### What is happening

**[Established]** A huge volume of commercial real-estate (CRE) loans — roughly \$875–\$936bn, about 17% of the \$5tn US CRE debt market — comes due in 2026, much of it written in 2019–2022 at 3–4% rates and maturing into a 6–8% world, against property values that are in many cases 16–35% lower. (The 2026 figure is actually down about 9% from 2025’s ~\$957bn, so the very peak may be passing.)

**[Established]** Office is the epicentre: the CMBS office delinquency rate hit a record 12.34% in January 2026. Banks hold roughly \$1.89tn of CRE loans on income-producing property, with securitised lenders holding another ~\$753bn. The long-running “extend and pretend” tactic — modifying and extending loans rather than recognising losses — is starting to break as lenders finally acknowledge losses.

**[Single-source]** One industry analysis estimates regional banks hold about \$396bn of the maturing debt and that over half exceed the 300% CRE-to-capital level regulators watch; **[Established]** regulators are tightening oversight of CRE-heavy banks, and the Fed’s loan-officer survey shows banks have tightened CRE standards for eight straight quarters.

### Why it is under-covered

“Office buildings in trouble” sounds like a niche real-estate problem. The under-covered angle is the link to regional banks — which hold a disproportionate share of this debt — and therefore to local lending, small-business credit, and the deposits of ordinary people who never think about CMBS.

### The debate

**The alarmist case:** a slow-motion solvency drain on regional banks as extensions expire, values stay depressed, and losses crystallise — echoing the regional-bank scare of 2023.

**The reassuring case:** this wall is the most widely anticipated in history, so markets and banks have been provisioning and adapting (extensions, recapitalisations, private-credit takeouts); CRE loans carry less leverage and more cushion than 2008-era mortgages, and bank CRE delinquencies remain far below the ~9% peak of 2010.

**Both sides accept:** office values have fallen hard, a large slug of debt must be refinanced into much higher rates, and regional banks are the concentrated pressure point.

### Stress test (the benign case)

Because everyone sees this coming, it may resolve as a grind rather than a crash: loan quality matters more than maturity volume, well-located properties with strong tenants refinance fine, private credit fills gaps, and the maturity peak is already easing from 2025. Insured depositors are protected

regardless. The vulnerability is concentration and confidence: a few regional banks with heavy office exposure and thin capital could still fail, and — as 2023 showed — fear can move faster than fundamentals once a name is in the spotlight.

### Historical parallel

The early-1990s savings-and-loan and CRE bust is the template for a drawn-out workout that takes years and shuts weak lenders without sinking the system. 2008 is the scarier comparison, but most analysts argue CRE today has lower leverage, more diverse funding, and far lower delinquencies than residential mortgages did then — a slow bleed, not a sudden break.

### In plain terms

Lots of loans on office buildings and other commercial property were taken out when interest rates were very low. They're now coming due all at once, but rates are much higher and many of those buildings are worth far less — especially offices. Smaller “regional” banks made a lot of these loans, so the strain lands on them. Your insured deposits are safe, but if local banks are nursing property losses, they lend less, which can squeeze small businesses and local economies. It's more likely to be a slow grind than a sudden crash, partly because everyone has seen it coming.

### What you can do now

**Stay within deposit-insurance limits.** In the US, FDIC insurance covers \$250,000 per depositor, per bank, per ownership category. If you're near that at one bank, spreading across banks (or using insured-sweep options) keeps you fully covered no matter what.

**Know your bank.** You don't need to panic-switch, but it's reasonable to be aware if your bank is unusually concentrated in commercial real estate; bank financial summaries are public.

**Keep some liquidity outside any single institution** so an isolated bank problem never freezes your access to cash.

**If you own affected property,** engage lenders early on extensions or restructuring — the worst outcomes hit those who wait until maturity day.

### What to watch

- CMBS office delinquency and special-servicing rates — the clearest distress gauge.
- Regional-bank loan-loss provisions, CRE concentration ratios, and any rating downgrades.
- The pace of “extend and pretend” ending — rising recognised losses vs continued extensions.
- Private-credit appetite to refinance maturing CRE — the relief valve.



## 8. The hidden consumer-debt squeeze: subprime cars and “phantom” instalments

CURRENT THREAT LEVEL: **5/10 — Elevated** *Acute at the bottom, contained overall*

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**Why this level:** Record subprime-auto stress and invisible BNPL debt at the fragile end, but small relative to the system with no clear broad spillover yet. **Raises it:** Rising unemployment; BNPL/auto delinquencies bleeding into cards and mortgages; a student-loan default surge. **Lowers it:** Stable employment and real wages; BNPL data reaching the credit bureaus; delinquencies plateauing.

**WHO THIS HITS:** *Anyone with a car loan, anyone who uses Buy Now Pay Later, parents of Gen Z, and anyone applying for a mortgage near someone who does.*

### What is happening

**[Established]** Two pockets of American household stress are flashing red while the averages look calm. First, subprime auto: 60-plus-day delinquencies among subprime borrowers hit their highest level in 32 years in January 2026, repossessions rose 43% between 2022 and 2024, and auto-loan rejection rates have roughly doubled. Loan terms have stretched to 84 months and negative equity (owing more than the car is worth) is at records.

**[Established]** Second, “Buy Now, Pay Later” (BNPL) — instalment plans from Klarna, Affirm, Afterpay and others — which regulators (the Richmond Fed) literally call “phantom debt” because much of it isn’t reported to the main credit bureaus. A borrower can stack loans across several apps and still look debt-free on a credit report. Because BNPL balances are small, people prioritise paying them — so their larger debts (cards, car, student loans) default first, meaning official metrics understate the strain.

**[Contested]** BNPL’s size is genuinely murky and reported numbers conflict. The Richmond Fed estimates real BNPL transaction value around \$70bn in 2025 (~1.1% of card spending) — small. Other sources cite annual purchase volume (GMV) near \$560bn and outstanding balances above \$400bn. These measure different things (a year’s spending vs money currently owed), which is exactly why the risk is hard to size. **[Single-source]** Separately, one analysis reports US student-loan defaults reappearing sharply — around a million in late 2025 and 2.6 million in early 2026 — as pandemic-era forbearance ends.

### Why it is under-covered

Aggregate consumer numbers look fine — overall credit-card delinquency is stable, real incomes edged up. That average masks a sharply “K-shaped” reality where the bottom tier is in deep stress, and a whole category of borrowing (BNPL) is partly invisible to the data everyone relies on. As one investor put it, there are two possible bubbles right now and only the AI one gets the headlines.

### The debate

**The alarmists argue:** this is the subprime-mortgage playbook — risky consumer debt, sliced up and partly invisible — with record auto delinquencies and hidden BNPL stacking pointing to a coming reckoning.

**The Fed and regulators counter:** BNPL is still small relative to credit cards; much of the “record” subprime auto delinquency reflects credit-score migration (during the pandemic, many borrowers’ scores rose, shrinking the subprime pool and flattering its delinquency math); and there’s no clear evidence yet of broad systemic spillover.

**Both sides accept:** the most financially fragile households are genuinely strained, BNPL data gaps are real and growing, and these are early-warning lights even if not yet a system-wide fire.

### Stress test (the benign case)

The reassuring read is solid: BNPL charge-off rates have actually been low (under 2% in recent reporting), credit-card delinquencies are near pre-pandemic norms, real median wages rose faster than inflation in early 2026, and the subprime-auto “record” is partly a statistical artifact. The risk to that case is the hidden layer — if rising unemployment meets phantom debt and the end of student-loan forbearance, stress can accelerate quickly and show up in the larger loans first, where it does real damage.

### Historical parallel

2008 subprime is the cautionary tale invoked most — invisible, fragmented, under-measured risk — though the scale here is far smaller and the systemic plumbing different. A fairer frame is the “K-shaped” consumer of recent cycles: averages hide a widening gap, and the people at the bottom run out of buffers long before the headline data admits it.

### In plain terms

On paper, American consumers look okay. But beneath the average, the people with the weakest finances are struggling: car-loan defaults among shaky borrowers just hit a 32-year high, and repossessions have jumped. At the same time, “pay in 4” instalment apps have let people pile up small debts that mostly don’t show on credit reports — so lenders (and the official statistics) can’t see them. People keep these small instalments current and let bigger bills slide, which hides how stretched they are. It’s not a national crisis yet, but it’s a warning light — and it’s easy to fall into personally without noticing.

### What you can do now

**Track every BNPL plan you have** in one place (a note or spreadsheet). The danger of “pay in 4” is that four small plans across four apps add up to a payment you can’t make.

**Treat BNPL as debt, not a discount.** If you couldn’t buy it with cash or on a card you’ll clear this month, pause before splitting it into instalments.

**Don’t roll negative equity into a new car loan.** Wrapping what you still owe on an old car into a new, longer loan is how people get deeply “underwater.” Keep cars longer; aim for loans of 60 months or less.

**Build a small emergency fund first** (even \$500–\$1,000). It’s the buffer that stops a surprise bill from becoming a late payment or a new instalment plan.

**If you have federal student loans,** confirm your repayment status now that forbearance has ended — don’t let a missed payment quietly become a default on your credit file.

### What to watch

- Affirm/Klarna delinquency and charge-off disclosures — the clearest window into the invisible layer.

- Whether credit-card and auto delinquencies rise even as BNPL absorbs spending — a sign underlying stress is worse than any single metric.
- Progress on BNPL data flowing to credit bureaus (FICO's BNPL-inclusive scores).
- Student-loan default figures and the unemployment rate — the accelerants.





## 9. The quiet retreat of insurance — and why it threatens your home's value

CURRENT THREAT LEVEL: **5/10 — Elevated** *Building, with household-level damage now*

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**Why this level:** Premiums and non-renewals are rising and the value-and-mortgage channel is real, but the systemic transmission is still gradual. **Raises it:** A major catastrophe season; accelerating insurer exits; home-value declines in high-risk zones. **Lowers it:** Successful mitigation and pricing reforms; reinsurance capacity returning; credible policy backstops.

**WHO THIS HITS:** *Every homeowner and renter — especially in disaster-exposed areas — plus anyone with a mortgage or hoping to buy.*

### What is happening

**[Established]** Property insurers are raising premiums sharply, declining to renew policies, and exiting whole regions as climate-driven disaster losses mount — not just in California and Florida, but in places like New England that aren't thought of as high-risk. US insured catastrophe losses reached \$112.7bn in 2024, up 36% year-on-year. A Consumer Federation analysis found roughly 1 in 13 US homeowners now uninsured, leaving about \$1.6tn in home value unprotected.

**[Established]** This is becoming a financial-stability issue, not just a household-budget one. Because mortgage lenders require insurance, when coverage becomes unaffordable or unavailable, home sales fall through, values drop, and — as the Dallas Fed has documented — mortgage delinquencies and defaults rise. A US Senate Budget Committee investigation described the trajectory as “2008 all over again,” except potentially more permanent: a flooded-out or uninsurable home may never recover its value.

**[Single-source]** Supporting figures to treat as leads: a 2026 survey found nearly half of home buyers/sellers hit insurance problems and 21% had a deal fall through; in Louisiana, 30–40% of mortgage applications reportedly fail over insurance costs; premiums rose around 20% over two years. **[Established]** State “FAIR plans” — insurers of last resort — now operate in 34 states plus DC, and are absorbing more risk as private insurers pull back.

### Why it is under-covered

It's reported as a personal-finance gripe — “my premium went up” — rather than as a systemic threat to the housing market and the financial system. The mechanism that turns rising premiums into falling home values and mortgage defaults is the under-told part.

### The debate

**The financial-stability camp argues:** this is a slow-building 2008-style risk — uninsurable homes lose value, mortgages sour, and losses ripple through lenders and local tax bases.

**The market-signal camp argues:** rising premiums are climate risk being priced correctly; suppressing prices or over-relying on state insurers of last resort just hides the risk and pushes the cost onto everyone else.

**Both sides accept:** premiums are rising fast, availability is shrinking, more homes are uninsured, and the mortgage market is exposed wherever insurance becomes unavailable.

### Stress test (the benign case)

Insurance markets adapt: new pricing models, parametric products, reinsurance, mitigation incentives, and stronger building codes can restore coverage in many areas, and most of the country is not at the extreme tail of risk. The vulnerability is that the worst-hit areas may be genuinely, permanently uninsurable, and that the damage concentrates on lower-income and minority households least able to absorb it — turning a market adjustment into a localized wealth shock.

### Historical parallel

The comparison everyone reaches for is 2008 — a housing-finance shock building quietly until it cascades. The crucial difference, and why some call it worse, is permanence: 2008 property values eventually rebounded, but a home that can't be insured against a recurring disaster may not. Observers also note the map of uninsurable areas can echo historical redlining, concentrating harm on the same communities.

#### In plain terms

Insuring a home is getting much more expensive, and in some places you can't get insurance at all, because disasters are causing record losses. This matters beyond your bill: you generally can't get a mortgage without insurance, so when coverage disappears, buyers vanish, home values fall, and more people miss mortgage payments. Experts warn this could hit the housing market the way the 2008 crisis did — but possibly more lastingly, because an uninsurable home may never regain its value. It's spreading beyond the obvious places like Florida and California.

#### What you can do now

**Don't go uninsured to save money.** An uninsured disaster can wipe out your largest asset. If premiums are painful, first compare quotes widely — prices vary a lot between insurers.

**Raise your deductible thoughtfully** to lower premiums if (and only if) you have the savings to cover that deductible after a loss.

**Invest in mitigation — it can cut premiums and keep you insurable.** Roof upgrades, defensible space against wildfire, storm shutters, and water-leak sensors are often rewarded by insurers and sometimes subsidised by state programs.

**Document your home now.** Photos/video of every room and major possession, stored in the cloud, make claims far easier and fairer after a disaster.

**Before buying a home,** get an insurance quote before you commit — not after — and check the area's disaster history and whether insurers are still writing new policies there. Know whether a state FAIR plan is your only option.

**Review your coverage limits** against today's rebuilding costs; many policies are under-insured after years of construction inflation.

### What to watch

- Insurer non-renewal and market-exit announcements by state; growth in FAIR-plan enrollment.
- The Dallas Fed / academic data linking insurance costs to mortgage delinquencies.
- Home-value softening in high-risk zip codes — the early sign of the value channel biting.

- Federal or state policy responses (backstops, mitigation funding, disclosure rules).



## Section 3 — Sovereigns & the Monetary Order

This section moves up a level, from private balance sheets to public ones and to the monetary system itself. Developing-world debt distress, France's slow fiscal-political fragility, central banks' steady accumulation of gold and gradual move away from the dollar, and the arithmetic of underfunded pensions are all stories about the solvency and credibility of the state and the money it issues.

They connect through trust: a sovereign's ability to borrow, a currency's role as a reserve, and a pension promise all rest on the belief that future taxpayers will honor today's commitments. When that belief erodes — through too much debt, too little growth, or too many promises — the adjustment shows up as higher yields, weaker currencies, or broken promises in the form of default, inflation, or benefit cuts. De-dollarization and gold-buying are the reserve manager's hedge against exactly this loss of faith. This is also the layer that constrains every other section: when sovereigns lose fiscal space, the backstops that contained funding and credit crises in the past become politically and financially harder to deploy.

## 10. The silent debt crisis in the developing world

**WHO THIS HITS:** *Mainly people in developing countries (through austerity and lost services); for everyone else it's a moral, migration, and geopolitical story — and a risk inside emerging-market funds.*

**CURRENT THREAT LEVEL:** **5/10 — Elevated** *High human toll, lower systemic spillover*

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**Why this level:** A genuine, ongoing crisis for billions, but fragmented across many small economies and walled off from rich-world markets, so the global-financial threat is moderate even as the human cost is high. **Raises it:** A strong-dollar surge or commodity shock tipping several countries at once; stalled restructurings. **Lowers it:** Dollar and real-yield easing; faster G20 Common Framework debt relief; commodity-price relief.

### What is happening

**[Established]** As of March 2026, 75 of 119 low- and middle-income countries assessed by the IMF and rating agencies are in or at risk of debt distress. UN figures put annual developing-country debt servicing near \$1.4tn, and more than 3.4 billion people — close to half of humanity — live in countries that spend more on debt interest than on health or education. Interest already exceeds 10% of government revenue in 56 countries, and 20% in 17 of them, a threshold strongly associated with default.

**[Established]** Africa is the epicentre. Restructurings (Zambia, Ghana and others) have been slow and partial; leaders argue the credit-ratings system punishes reform. A recent report found Nigeria's debt payments running near five times its combined health and education spending. The 2026 Middle East energy shock worsens the squeeze for fuel importers, and a faith-led "Turn Debt into Hope" Jubilee campaign has revived calls for relief.

### Why it is under-covered

The UN Secretary-General named the exact reason: because the crisis is concentrated in poor countries, it is "not judged to pose a systemic risk to the global financial system" — so markets, and therefore most financial media, look away. It is a humanitarian and geopolitical catastrophe that is invisible precisely because it does not threaten rich-world portfolios.

### The debate

**Creditors and the IMF argue:** much of the debt is sustainable after fiscal adjustment; lending and reform programmes are the path back to market access.

**Critics argue:** the IMF functions as "the world's debt collector," its prescribed measures (e.g. VAT increases) are regressive, and the architecture forces a choice between servicing creditors and funding development.

**Both sides accept:** servicing burdens are at multi-decade highs, borrowing costs for these states run several times those of the US or Germany, and the diversion from health, education and investment is real and growing.

### Stress test (why it may not blow up)

There is a coherent case that this stays a slow burn rather than a 1980s-style cascade: exposures are fragmented across many small economies and many creditors (now including China and private bondholders), so there is no single domino. Multilateral backstops, the G20 Common Framework, and case-by-case restructurings can keep most countries limping along. But "no cascade" is not "no crisis" —

the human and developmental cost compounds quietly, and a synchronised dollar surge or commodity shock could still tip several at once.

### Historical parallel

The 1980s Latin American debt crisis — the “lost decade” — is the template: foreign-currency debt plus a strong dollar and high global rates produced a decade of austerity, lost growth and political upheaval, ultimately resolved only by the Brady Plan’s debt write-downs years later. The lesson is that delay is itself a policy with a body count, and resolution tends to come late and via principal reduction, not endless rollovers.

### In plain terms

Dozens of poorer countries now spend more on debt interest than on health or education, squeezing billions of people. It’s a real, ongoing crisis — but because it’s concentrated in poor countries and doesn’t threaten rich-world markets, it gets ignored. It’s unlikely to trigger a global financial crisis, yet it’s a slow humanitarian disaster and a quiet driver of migration and instability.

### What you can do now

**This one is mostly awareness and values**, not a portfolio move — but it’s worth understanding as a force behind global migration and politics.

**Watch the US dollar.** A stronger dollar quietly makes this worse for many countries at once — a useful signal even for your own emerging-market exposure.

**Know what your EM bond funds hold.** If you invest in emerging-market debt, you’re exposed to these dynamics.

**If you care about the issue**, debt-relief and development charities and campaigns are where individuals can plug in.

### What to watch

- The US dollar index (DXY) and US real yields: a stronger dollar mechanically raises the local-currency cost of foreign-currency debt.
- Sovereign foreign-currency bond spreads for frontier issuers, and the count of new IMF programmes or missed coupons.
- Progress (or paralysis) in G20 Common Framework restructurings — the test of whether the system can resolve rather than defer.
- Commodity and food prices: importers’ fiscal space lives and dies here.





## 11. France: the eurozone's quiet structural fault line

CURRENT THREAT LEVEL: **4/10 — Moderate** *Acute phase eased, structure fragile*

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**Why this level:** February's budget passage calmed spreads and the ECB backstop deters attacks, but the structural deficit and political fragility remain. **Raises it:** Snap elections or a government collapse; a credit-rating downgrade; the 5% deficit target slipping. **Lowers it:** Stable governance; credible consolidation; spreads holding near pre-2024 levels.

**WHO THIS HITS:** Europeans (via bonds, banks, and the euro), and anyone globally with European exposure in their pension or funds — plus a lesson in how political gridlock becomes a financial cost.

### What is happening

**[Established]** After nearly two years of paralysis since President Macron's 2024 snap election produced a hung parliament, France finally adopted a 2026 budget on 2 February 2026 — pushed through under Article 49.3, surviving two no-confidence votes only because Socialist lawmakers declined to bring the government down. The saga cost two prime ministers their jobs before Sébastien Lecornu's fragile minority government got it over the line.

**[Established]** The budget targets a 5% deficit for 2026 (down from 5.4% in 2025); the original 4.6% target was abandoned after pension reform was frozen. Public debt sits near 114–116% of GDP — the eurozone's third-highest after Greece and Italy — and is projected to climb toward ~125% by 2029. France remains under the EU's excessive-deficit procedure.

**[Established]** The market's verdict is the tell. The spread between French (OAT) and German (Bund) 10-year yields widened at its peak to levels on par with the 2012 eurozone crisis, and at one point French borrowing costs matched or exceeded Italy's — a once-unthinkable milestone. After February's budget passage, the risk premium eased back toward mid-2024 levels, but the structural fragility did not go away.

### Why it is under-covered (globally)

Outside Europe it reads as domestic French politics. But France is the eurozone's second-largest economy, and roughly 55% of its government bonds are held by foreign investors — the group most sensitive to bad news. When France wobbles, it is a euro-area problem, not just a Paris one.

### The debate

**The worriers argue:** serial government collapses, a structural deficit, frozen pension reform, and deteriorating demographics mean France can't credibly stabilise its debt; each crisis ratchets borrowing costs higher.

**The relaxed argue:** France still borrows cheaply by historical standards, has a long average debt maturity (~9 years) that blunts rate rises, deep and liquid markets, and — crucially — the ECB's backstop (the Transmission Protection Instrument) standing behind it.

**Both sides accept:** the political system struggles to deliver consolidation, debt is rising, and the market now charges France a standing "political risk premium."

### Stress test (why it probably doesn't blow up soon)

The eurozone has something the 2012 crisis lacked: a credible central-bank backstop. The mere existence of the ECB's TPI deters speculative attacks, and France's long maturities mean higher rates feed through slowly. The real risk is not a sudden default but a slow erosion — a higher permanent interest bill that crowds out spending, plus tail scenarios (snap elections, an early end to Macron's term) that could reprice risk faster than the backstop calms it.

### Historical parallel

The 2010–12 eurozone sovereign-debt crisis is the obvious reference, but the ECB's post-2012 “whatever it takes” toolkit changes the ending. A sharper warning is the UK's September 2022 gilt crisis: a credible-looking advanced economy saw its bond market seize in days once confidence cracked — proof that “it can't happen here” is not a risk model.

### In plain terms

France keeps struggling to agree on a budget — the fight has toppled two prime ministers — because no party has a majority. Its government debt is high and rising, and for a moment investors treated France as riskily as Italy, which is striking for a core European country. A budget finally passed in February 2026, calming markets somewhat, but the underlying problem (spending more than it raises, with politics too divided to fix it) is still there. Because France is huge and its debt is widely held abroad, its troubles are really a euro-wide issue. A full crisis is unlikely soon mainly because the European Central Bank stands ready to step in.

### What you can do now

**For most non-Europeans, this is awareness, not action** — but it's worth checking whether your bond or “international” funds carry heavy French/eurozone government-debt exposure.

**Diversify geography.** If a large share of your savings is tied to one country or currency bloc, broadening across regions reduces single-country political risk.

**Watch the euro if you hold it or earn in it;** persistent political risk premiums can weigh on the currency over time, affecting purchasing power.

**Take the general lesson:** political gridlock has a real price (higher borrowing costs), which is a useful lens for assessing any government's finances, including your own country's.

### What to watch

- The OAT–Bund spread (and OAT vs Italian BTP) — the live fever chart of French risk.
- Any move toward snap elections or instability around Macron's presidency.
- Credit-rating actions and whether the 5% deficit target actually holds.
- Any signal the ECB might use (or be asked to use) its TPI backstop.



## 12. Central banks are quietly buying gold and trimming the dollar

CURRENT THREAT LEVEL: **3/10 — Moderate** *Structural signal, not a near-term threat*

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**Why this level:** A real, multi-decade reserve shift, but slow-moving with no imminent trigger; more a confidence signal than an acute risk. **Raises it:** A new reserve-weaponisation event; a disorderly dollar decline; faster non-dollar trade settlement. **Lowers it:** A stable dollar reserve share; orderly diversification; renewed official demand for Treasuries.

**WHO THIS HITS:** *Every saver — because it's a signal about long-run confidence in the dollar and government debt, and it shapes the value of money you hold.*

### What is happening

**[Established]** The world's central banks have bought more than 4,000 tonnes of gold between 2022 and early 2026 — the largest sustained official accumulation in modern history. They bought 863 tonnes in 2025 (down from ~1,090 in 2024 only because record prices made them more selective, not less committed). Gold hit an all-time high of \$5,589/oz on 28 January 2026 before easing to around \$4,300 by mid-June.

**[Established]** Meanwhile the US dollar's share of global currency reserves has fallen to roughly 56.8% (late 2025) from about 72% in the early 2000s — the lowest since 1994 — a steady, multi-decade drift that has continued even when the dollar was strong. The 2026 World Gold Council survey (a record 76 central banks) found 45% plan to add gold and 74% expect the dollar's reserve share to fall further.

**[Single-source]** Strikingly, Morgan Stanley research notes gold now makes up a larger share of central-bank reserves than US Treasuries for the first time since 1996. **[Established]** Much of the buying is under-reported — China discloses only small amounts even as its gold imports run far higher — so official tallies understate the trend.

### Why it is under-covered

Reserve management is the dullest-sounding topic in finance, and “gold price hits record” gets covered as a market story, not a monetary one. The real signal — the people who manage national wealth quietly reducing dollar dependence and removing counterparty risk — is a slow architectural shift that rarely makes front pages until it's far advanced.

### The debate

**The “de-dollarization” camp argues:** this is a structural turn away from a dollar that has been “weaponised” through sanctions; gold can't be frozen or inflated by another government, so sovereigns are buying insurance against US financial power and debt.

**The “just diversification” camp (incl. Fed research) argues:** the dollar still dwarfs all rivals at ~57% of reserves, no currency can replace it, and gold buying is prudent hedging — not the end of the dollar system.

**Both sides accept:** the dollar's reserve share has fallen for 25 years, central-bank gold demand is at historic highs, and no single currency is filling the gap the dollar is vacating.

### Stress test (the benign case)

Nothing here implies a dollar collapse. The dollar remains dominant in trade invoicing, debt issuance, and payments; reserve shifts unfold over decades; and a 57% share is still overwhelming. Gold buying can simply be sensible diversification after a generation of over-concentration. The caution is directional: reserve-currency transitions historically show up in the data long before they register in public consciousness, and a steady bid for gold over Treasuries is the same “who will buy government debt?” question from Set 1, seen from the official-sector side.

### Historical parallel

The sterling-to-dollar handover across the first half of the 20th century is the model for how slowly reserve status changes — measured in decades, not quarters. The 1970s breakdown of the gold-anchored Bretton Woods system is the reminder that monetary regimes do shift, and that gold tends to be where capital hides when faith in paper claims wavers.

#### In plain terms

The institutions that manage countries' savings — central banks — have been buying gold at the fastest pace in modern history and slowly holding fewer dollars. They started in earnest after the US froze Russia's reserves in 2022, which taught everyone that dollars held abroad can be switched off. This doesn't mean the dollar is finished — it's still by far the most-used currency — but it's a quiet vote of reduced confidence in holding everything in dollars and government IOUs. For an ordinary saver, it's a signal worth noting about the long-run value of cash and debt, and why a little diversification can make sense.

#### What you can do now

**Understand gold's role before buying any.** It's a diversifier and a hedge against currency and crisis risk — not a guaranteed money-maker. It pays no income and can fall sharply (it already dropped from ~\$5,600 to ~\$4,300 in months).

**Keep any allocation modest.** Many advisers think of single-digit percentages of a portfolio; concentrating heavily in gold is its own risk.

**Beware markups and scams.** Physical-gold dealers can charge large premiums, and “rare coin” pitches are a classic trap. Low-cost, well-known gold funds avoid storage and authenticity headaches; if you want physical, compare premiums and use reputable dealers.

**Take the broader hint:** diversifying across currencies and real assets is a reasonable response to long-run concerns about debt and inflation — gold is one option among several, not a silver bullet.

### What to watch

- The IMF COFER data on the dollar's reserve share, released quarterly.
- World Gold Council quarterly demand reports and the annual central-bank survey.
- Whether large holders (especially China) report more of their actual buying.
- Any moves toward non-dollar trade settlement among major economies.



### 13. The retirement math nobody wants to say out loud

CURRENT THREAT LEVEL: **4/10 — Moderate** *Large but distant*

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**Why this level:** A near-certain, very large shortfall — but dated to the 2030s–2050s with no near-term trigger; the risk is political delay, not imminent collapse. **Raises it:** Abrupt benefit-cut or trust-fund headlines; a pension-fund leverage event (cf. UK 2022); reform-driven instability. **Lowers it:** Early, gradual reforms; productivity and immigration gains; rising funded ratios.

**WHO THIS HITS:** *Everyone who hopes to retire — especially younger workers and anyone relying on a state pension or Social Security.*

#### What is happening

**[Established]** Populations are ageing faster than retirement systems were designed for. Across the OECD there are about 33 people aged 65+ for every 100 working-age adults today, heading toward 52 by 2060. In the US, the number of workers per retiree fell from 5.1 in 1960 to a projected ~2.1 by 2030. Pay-as-you-go systems — where today's workers fund today's retirees — strain as that ratio falls.

**[Forecast]** The World Economic Forum estimates the six largest pension systems face a combined retirement-savings gap of about \$224tn by 2050, with the US alone accounting for ~\$137tn (over 60%), and reckons the global gap widens by roughly \$28bn every day. **[Established]** Global pension assets across 22 major markets hit a record \$68.3tn at end-2025 — large, but the six biggest systems hold only around 25 cents for every dollar they are projected to owe by 2050.

**[Established]** Concrete near-term example: the US Social Security retirement trust fund is projected to be depleted around 2033, after which incoming taxes would cover only about 77% of scheduled benefits — an automatic ~23% cut absent reform, roughly \$18,100 a year for an average retiring couple. US state and local pensions carry unfunded obligations estimated above \$1.2tn.

#### Why it is under-covered

It's the ultimate slow-motion story — a problem dated to the 2030s–2050s that no politician is rewarded for fixing today, because every fix (later retirement, higher contributions, lower benefits) is unpopular. So it's perpetually deferred, and coverage spikes only around flashpoints.

#### The debate

**The crisis camp argues:** the arithmetic is unforgiving — the WEF likens it to “the financial equivalent of climate change” — and decades of postponement mean the eventual adjustments (benefit cuts, tax rises, later retirement) will be abrupt and painful.

**The manageable camp argues:** gradual, well-designed reforms — raising retirement ages in line with longevity, higher contributions, immigration, productivity growth, and broader workforce participation — can close much of the gap if started early; many systems are adjusting.

**Both sides accept:** dependency ratios are deteriorating, pre-funding is insufficient, and reform is politically toxic — which is why it keeps being delayed.

#### Stress test (the benign case)

This is a known, forecastable, gradual problem — which is exactly why it's solvable. Countries that started early (raising retirement ages, building funded pillars) have stabilised their systems; productivity

gains (including from AI) could lift the wage base; and immigration can soften dependency ratios. The risk is political, not arithmetic: reforms that destabilise governments get watered down, and gaps compound while leaders postpone — so the burden lands hardest on those who didn't save privately to fill it.

### Historical parallel

Two warnings. Politically, France's 2023 attempt to raise the retirement age nearly toppled the government, and the UK's 2022 pension-fund (LDI) blow-up showed even "well-funded" systems can fracture — both in countries considered relatively healthy. Structurally, Japan is the live preview: the most-aged major economy, already managing decades of demographic decline, shows both the strain and the slow adaptations the rest of the world is heading into.

### In plain terms

People are living longer and having fewer children, so there are fewer workers supporting each retiree. The pension and Social Security systems built for a younger world are short of money — by trillions, over the coming decades. It's not a crisis tomorrow, but the longer governments wait, the harsher the eventual fix: later retirement ages, higher taxes, or smaller benefits. In the US specifically, Social Security's main fund is on track to run low around 2033, which would automatically trim benefits by roughly a quarter unless Congress acts. The practical takeaway: assume you may get somewhat less, somewhat later, than promised — and plan for it.

### What you can do now

**Read your annual statement.** In the US, your Social Security statement (at [ssa.gov](https://ssa.gov)) shows your projected benefit; check it. Workers elsewhere should locate their state-pension forecast. Knowing the number is step one.

**Capture every bit of free money.** If an employer matches retirement contributions, contributing at least enough to get the full match is the closest thing to free money in finance.

**Start early and automate.** Because of compounding, money saved in your 20s–30s does far more work than the same amount later. Automatic monthly contributions beat willpower.

**Plan for "less, later."** It's prudent to assume retirement ages may rise and public benefits may be trimmed; building your own savings buffer protects you if they are.

**Know what kind of pension you have.** A "defined-benefit" pension promises a set payout (employer bears the risk); a "defined-contribution" plan (like a 401(k)) does not — you do. That changes how much you need to save yourself.

**Diversify your retirement income** across sources (state benefit, workplace plan, personal savings) so no single shortfall sinks your plans.

### What to watch

- Official trust-fund depletion dates (e.g. the annual US Social Security Trustees report).
- Retirement-age and contribution reforms — and the political turbulence around them.
- Pension-fund funding ratios (public and corporate) and any leverage strategies (the UK 2022 lesson).



- Birth-rate and immigration trends — the long-run drivers of the dependency ratio.



## Section 4 — Geoeconomics & Trade Weapons

These four stories are about interdependence turned into leverage — the deliberate use of economic chokepoints as instruments of state power. China's export of deflation, the weaponization of rare earths and critical minerals, the single-company lithography chokepoint that gates advanced chips, and the contested sea lanes and undersea cables that physically carry trade and data are all nodes where one actor can squeeze the whole system.

The connective tissue is concentration: each is a point where global supply runs through a needle's eye — one country's overcapacity, one country's refining monopoly, one firm's machines, one strait's tankers. That concentration is efficient in peacetime and coercive in conflict. The 2026 Gulf disruption made the sea-lane risk concrete; the rare-earth and lithography stories show the same logic applied to inputs the digital and defense economies cannot substitute quickly. These risks move more slowly than a funding run but are harder to reverse, because rebuilding a supply chain takes years where a margin call takes minutes — and a trade shock here is often the exogenous spark that lights the leverage and credit fires elsewhere.

## 14. China is exporting its deflation — the global transmission nobody frames clearly

**WHO THIS HITS:** Anyone who buys goods, works in manufacturing, or invests — China exporting cheap goods and deflation reshapes prices, jobs, and trade politics worldwide.

**CURRENT THREAT LEVEL:** **6/10 — Elevated** *Entrenched and radiating outward*

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**Why this level:** Persistent, multi-year deflation with an export surge that is straining trading partners; structural and slowly worsening rather than acute. **Raises it:** A managed yuan depreciation; an escalating tariff/anti-dumping cycle; a local-government or bank stress event. **Lowers it:** Genuine consumption stimulus; producer prices turning positive; a negotiated trade de-escalation.

### What is happening

**[Established]** China has now run roughly ten consecutive quarters of deflationary pressure — the longest stretch since its market transition in the late 1970s. Producer prices have been negative for over 30 months; consumer inflation averaged about 0% in 2025. The cause is a structural demand deficiency: a multi-year property crash (a fifth year of contraction, with tens of millions of vacant homes) has crushed household wealth and confidence, while state-directed investment keeps adding industrial capacity.

**[Established]** With demand weak at home, that capacity is sold abroad: China posted a record trade surplus near \$1.2tn in 2025. The IMF's Article IV assessment is explicit that high export volumes are creating adverse spillovers, and a US Congressional Research Service summary cites an estimate that the renminbi is undervalued by around 16%. Beijing set its 2026 growth target at 4.5–5%, the lowest in decades.

**[Single-source]** One account puts the local-government debt pile near \$18.9tn against land sales that once supplied 40% of local revenue; **[Established]** the broader point — a doubling of total debt to over 300% of GDP and a local-government financing overhang that forces banks to roll loans to “zombie” firms — is well supported and entrenches the overcapacity.

### Why it is under-covered (as a global story)

China's deflation is covered — as a China story. What is under-framed is the transmission: a \$20tn economy quietly exporting disinflation and excess supply into global tradable goods reshapes prices, margins and politics in dozens of other countries, feeding tariff fights, squeezing emerging-market manufacturers, and complicating every other central bank's job.

### The debate

**The catastrophist view:** a debt-deflation spiral that ends in a balance-sheet crisis or a sharp devaluation.

**The stagnation view:** an “L-shaped” grind — low growth, persistent deflation, no collapse — Japanification with Chinese characteristics.

**Beijing's view:** risks are contained, and “new quality productive forces” (EVs, solar, robotics, AI) will replace property as the engine.

**All sides accept:** the deflation is real and persistent, the consumer-demand response has been modest relative to the problem, and the export surge is straining trading-partner relations.

### Stress test (the benign case)

China has confounded collapse calls for over a decade. It retains enormous policy room (rates, reserve requirements, special bonds), a closed capital account that limits flight, and genuine global leadership in EVs, solar and automation that could yet rebalance growth. Cheaper Chinese goods also lower input costs and inflation for importers — a benefit, not only a threat. The flaw in the benign case: every quarter of deferred demand-side reform deepens the debt-deflation loop and widens the external imbalance that provokes a protectionist backlash.

### Historical parallel

Japan after 1990 is the canonical comparison: a property and balance-sheet bust followed by decades of deflationary stagnation despite heavy investment. The cautionary twist is scale — Japan was a creditor exporting capital; China is a far larger share of global manufacturing exporting goods, so its domestic deflation radiates outward with greater force, much as the 1930s saw beggar-thy-neighbour dynamics when surplus capacity met weak demand.

#### In plain terms

China's economy is stuck in a slump where prices keep falling. To cope, it's producing and exporting huge volumes of cheap goods. For shoppers elsewhere that can mean lower prices; for factories and workers in other countries it means brutal competition, and it fuels tariffs and trade fights. It's a China problem that quietly becomes everyone's problem through trade.

#### What you can do now

**As a shopper, cheaper imports can help you** — but expect more 'trade war' headlines and tariffs that can raise some prices and rattle markets.

**If your job or industry competes with Chinese exports** (manufacturing, steel, solar, EVs), watch for pricing pressure and plan accordingly.

**For investors, diversify.** Broad emerging-market and commodity exposure is sensitive to China's slowdown.

**Don't assume low inflation is permanent.** Trade fights can flip cheap-goods disinflation into cost-push inflation quickly.

### What to watch

- China's PPI and the CNY exchange rate: a managed depreciation would intensify exported deflation and invite retaliation.
- Export volumes vs values, and tariff/anti-dumping actions by the EU, US and emerging markets — the political feedback loop.
- Any genuine consumption stimulus (social safety net, direct transfers) vs more supply-side investment — the signal of whether Beijing is treating the cause or the symptom.
- Local-government financing-vehicle restructurings and bank net interest margins.



## 15. Critical minerals as a weapon: “availability is not access”

**WHO THIS HITS:** Anyone who buys cars, electronics, or appliances, plus workers in manufacturing, clean energy, and defense — these minerals are in almost everything with a motor or magnet.

**CURRENT THREAT LEVEL:** **6/10 — Elevated** *Coercive tool on a fragile truce*

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**Why this level:** China’s processing chokepoint is real, the November 2025 pause is temporary with the April controls intact, and access can be throttled quickly. **Raises it:** Reinstated or expanded export controls; lapse of the pause; a geopolitical flashpoint over Taiwan or tech. **Lowers it:** An extended or broadened truce; non-China processing and magnet capacity coming online on schedule.

### What is happening

**[Established]** China controls roughly 90% of global rare-earth permanent-magnet production and about 91% of refining and separation — the processing chokepoint, not just the mines. Through 2025 it built a licensing regime into a coercive instrument: April 2025 controls on seven heavy rare earths and magnets; an October 2025 expansion adding five more elements and, via Announcement 61, an extraterritorial reach over foreign goods containing Chinese-origin material; then a November 2025 pause (Announcements 70/72) that suspended the October measures but preserved the April controls and the licensing machinery.

**[Established]** The key insight from trade data: availability is not access. China’s overall magnet exports rose even as shipments to the US fell sharply, because supply is selectively allocated through discretionary licensing. US yttrium imports ran 17 tons across April–December 2025 versus 333 tons in the prior period; aerospace makers have rationed material. Ford halted Explorer production for a week in May 2025; European magnet prices reached up to six times Chinese levels.

### Why it is under-covered

Headlines treat each tariff truce as “resolved.” The structural reality — that the November détente is a temporary, revocable pause sitting atop intact controls, and that the real bottleneck is processing know-how that takes years to replicate — gets lost between summit news cycles.

### The debate

**The hawk view:** decouple fast; subsidise an ex-China chain across mining, separation, metallisation and magnet-making at any cost, because the leverage is intolerable.

**The realist view:** diversification is possible but slow, capital-heavy and chemistry-bound; switching costs run into billions per firm and decoupling could cut first-year operating profits 15–50% in exposed firms.

**Beijing’s framing:** conservation and national security.

**All sides accept:** the chokepoint is real, processing (not ore) is the binding constraint, and meaningful displacement is measured in years.

### Stress test (the benign case)

Diversification is genuinely underway: a US–Australia critical-minerals framework, a layered US–Japan partnership (including deep-sea rare-earth mud research), nearly \$4bn in US EXIM letters of intent, and new magnet capacity coming online in summer 2026. Substitutes (iron-nitride magnets, magnet-free motor architectures) are advancing. The benign case says prices stay elevated but supply chains adapt

through triage and recycling. Its weakness: every credible analysis still puts true processing independence years away, and a single licensing freeze can outpace the build-out.

### Historical parallel

China's 2010 rare-earth embargo against Japan is the precedent: it spurred a wave of patenting and some ex-China capacity, but did not close the gap quickly — dependence persisted for years. The deeper analogue is the 1973–74 oil shock, which is exactly why the IEA has launched a Critical Minerals Security Programme modelled on the emergency-stockpile cooperation that followed the oil embargo. The lesson: chokepoint shocks accelerate diversification but reward the holder for a long interim.

### In plain terms

China controls about 90% of the processing for the 'rare earth' magnets used in EVs, electronics, wind turbines, and weapons. It has turned export licenses into a bargaining chip — supply can be quietly throttled even when the minerals technically exist. A November 2025 truce paused the harshest measures but left the controls in place, and building alternatives outside China takes years, so this leverage isn't going away soon.

### What you can do now

**Mostly awareness:** expect occasional shortages, price bumps, or delays in products that rely on these magnets (some cars, electronics) when controls tighten.

**If you're in an affected industry** (autos, electronics, defense, clean energy), supply-chain diversification and inventory planning matter.

**Treat single-stock 'rare earth' bets with the usual caution;** it's a long-running theme, not a guaranteed winner.

**No need to stockpile anything personally** — the impact shows up as prices and availability, not a household emergency.

### What to watch

- Chinese export-license approval times and volumes by element and destination — the real-time gauge of access vs availability.
- Whether the November pause is extended, expired, or reinstated (watch the late-2026 dates in the announcements).
- Heavy rare-earth prices (dysprosium, terbium, yttrium) and ex-China premiums.
- Commissioning milestones for non-China separation and magnet plants in 2026–2027 — announcements vs actual qualified output.





## 16. The lithography chokepoint: one machine the whole digital world needs

**WHO THIS HITS:** Everyone who uses any modern technology, and anyone invested in AI or chips — every advanced chip on Earth depends on one company's machines, made with one supplier's optics, in one country.

CURRENT THREAT LEVEL: **5/10 — Elevated** *Extreme single point of failure*

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**Why this level:** The most concentrated chokepoint in the entire economy — one company, one optics supplier, key fabs in Taiwan — though there is no imminent disruption. **Raises it:** A Taiwan crisis or an incident at ASML (Veldhoven) or Zeiss; a sharp escalation of export-control retaliation. **Lowers it:** Geographic diversification of advanced fabs (US, Japan, Europe) maturing; stable cross-strait and export-control conditions.

### What is happening

**[Established]** Every advanced chip in the world — every AI accelerator, every flagship phone processor — is made using extreme-ultraviolet (EUV) lithography machines built by exactly one company: ASML, in the Netherlands. ASML has a 100% monopoly on EUV and ~94% of lithography overall; Nikon and Canon exited advanced lithography years ago. ASML can build only ~50–60 of these machines a year — a physical ceiling on how much leading-edge chip capacity the world can add. Each next-generation High-NA machine costs around \$400 million.

**[Established]** The concentration runs deeper than one company. ASML itself depends on a single optics supplier — Germany's Zeiss — for the flawless mirrors EUV requires; the future pace of the technology is effectively gated by what Zeiss can build. The leading-edge chips themselves are made overwhelmingly by TSMC in Taiwan (which also depends on Taiwan's water, per Story 3). So the entire digital economy funnels through one machine-maker, one optics-maker, and one island.

**[Established]** This chokepoint is also a weapon. EUV machines have never been sold to China, and US-led export controls keep tightening; China's share of ASML sales fell from ~33% in 2025 toward ~19–20% in 2026. China is reportedly a decade or more behind on domestic EUV. **[Contested]** In April 2026 TSMC said it would stick with current-generation EUV through at least 2029 rather than buy High-NA immediately — a reminder that even the indispensable supplier has only one buyer big enough to absorb its bleeding edge, a mutual dependence that cuts both ways.

### Why it is under-covered

ASML is the most strategically important company almost no one outside the industry can name. The chokepoint is abstract (a lithography machine in Veldhoven) and has been stable for years, so it hides in plain sight — until a Taiwan scenario or a single-site incident makes its fragility suddenly concrete.

### The debate

**The fragility camp argues:** this is the ultimate single point of failure — one firm, one optics supplier, one island — and a disruption (war over Taiwan, an accident, escalating controls) would halt advanced-chip progress for the whole world.

**The resilience camp argues:** the very irreplaceability creates caution on all sides (no one benefits from breaking it), diversification is underway (new fabs in the US, Japan, and Europe), and older chips can carry much of the economy for a long time.

**Both sides accept:** the concentration is extreme and real, processing/manufacturing know-how takes decades to replicate, and meaningful diversification is years away.

### Stress test (the benign case)

Mutual dependence is stabilizing: ASML needs TSMC as much as TSMC needs ASML, China is deterred from a move on Taiwan partly because it would destroy the thing it wants, and a multi-year push is spreading advanced fabs across the US, Japan, and Europe. Older-node chips — which run most of the physical economy — are far less concentrated. The vulnerability: none of that helps in the tail event. A single bad day in the Taiwan Strait, in Veldhoven, or at Zeiss has no quick substitute, and “years away” diversification is no comfort on the day it matters.

### Historical parallel

The 2020–22 chip shortage — when a shortfall of even cheap, mature chips idled auto plants worldwide — was a preview of how a narrow bottleneck cascades through the real economy. The deeper analogue is the oil chokepoints of the 1970s: a single concentrated input, once disrupted, reorders the global economy and triggers a scramble for alternatives that takes a decade to bear fruit.

### In plain terms

Every cutting-edge computer chip — in AI systems, phones, everything — is made using a machine that only one company on Earth knows how to build: ASML, in the Netherlands. That company in turn depends on a single supplier (Zeiss in Germany) for its precision mirrors, and the chips themselves are mostly made by one company in Taiwan. So the entire digital world runs through one machine-maker, one optics-maker, and one island. It's been stable for years, and everyone has reasons not to break it — but it's the most concentrated single point of failure in the global economy, and a war over Taiwan or a serious accident would have no quick fix.

### What you can do now

**This is awareness, not an action item** — but it explains why ‘Taiwan risk’ is treated as a global-economy risk, not a regional one.

**Mind concentration in your investments.** AI and chip exposure ultimately funnels through a very small number of irreplaceable companies and one geography; broad diversification (across regions and sectors) is the realistic hedge for an ordinary investor.

**Expect tech-hardware price and availability swings** whenever export controls or cross-strait tensions escalate; big-ticket electronics purchases can be worth timing around obvious flashpoints.

**Don't try to trade the geopolitics.** Tail-risk timing is a fool's errand; a cash buffer and diversification protect you better than predicting a Taiwan crisis.

### What to watch

- Cross-strait tensions and any military escalation around Taiwan — the dominant tail risk.
- Export-control rounds and Chinese retaliation (and China's share of ASML sales).
- Progress and yields at new advanced fabs outside Taiwan (US, Japan, Europe) — real diversification vs announcements.
- ASML order backlog and High-NA adoption timelines — the health of the chokepoint itself.





## 17. Sea lanes and undersea cables: the physical arteries of globalization

**WHO THIS HITS:** Everyone — through fuel and goods prices, and through the internet itself. About 80% of trade moves by sea and ~95% of data moves under it, much of it through a few narrow, contested passages.

**CURRENT THREAT LEVEL:** **7/10 — High** *A live energy-and-trade crisis*

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**Why this level:** A live crisis: the Strait of Hormuz was closed for months, the Red Sea remains contested, undersea cables are openly threatened, and the IEA calls the resulting energy shock worse than the 1970s. **Raises it:** A breakdown of the Hormuz ceasefire; a major cable-cutting event; a second chokepoint (Bab el-Mandeb, Malacca, Panama) seizing at the same time. **Lowers it:** A durable Gulf ceasefire reopening Hormuz; restored Red Sea transit; cable redundancy and protection improving.

### What is happening

**[Established]** Globalization runs through a handful of narrow passages, and several are under stress at once. The Strait of Hormuz — through which roughly a fifth of the world's traded oil moves — was effectively closed for months in 2026 amid the US-Iran conflict, stranding up to ~20,000 seafarers on ~2,000 vessels; the IEA called the resulting oil-and-gas shock more severe than the crises of 1973, 1979, and 2022 combined. A US-Iran ceasefire is now slowly reopening it, but the situation is fragile, and on 8 June 2026 the Houthis declared a renewed ban on Israeli ships in the Red Sea.

**[Established]** The same chokepoints are also **digital** chokepoints. Around 95% of intercontinental data travels through undersea cables, and they cluster through the same narrow waters: roughly 90% of Europe-Asia cables run through the Red Sea, and multiple major systems thread the Strait of Hormuz. In 2024, cable cuts in the Red Sea (one caused by a ship's dragged anchor) disrupted an estimated 25% of Europe-Asia internet traffic. In 2026, Iran-linked media pointedly highlighted the Hormuz cables' vulnerability, and Meta paused work on a major Gulf cable system. Unlike ships, which can reroute over weeks, cable outages hit financial trading, cloud services, and communications within hours — and repairs can take weeks to months.

**[Established]** Other chokepoints are flashing too. The Panama Canal cut transits ~40% during the 2023-24 drought (a water link to Story 3), and in 2026 became a US-China flashpoint after Panama's top court voided a Hong Kong company's port contracts — prompting China to detain scores of Panama-flagged vessels. The Strait of Malacca (80,000+ vessels a year) saw Indonesia float a transit toll. The deeper shift: disruption no longer needs a formal blockade — drones, mines, cyberattacks, dragged anchors, and insurance spikes can achieve the same effect far more cheaply.

### Why it is under-covered

The oil-tanker angle gets covered; the cable angle almost never does, even though a coordinated cable disruption could hit the economy as hard as closing a canal. These are the literal arteries of trade and data, owned mostly by private firms, governed by weak international frameworks, and invisible until they're cut.

### The debate

**The fragility camp argues:** the world rebuilt itself around just-in-time, open sea lanes and a few cable corridors, and that assumption is now broken — cheap, deniable attacks can choke trade, energy, and data at multiple points at once.

**The resilience camp argues:** shipping reroutes (around the Cape, as it did for the Red Sea), cables get repaired and rerouted, and markets adjust — painfully but without collapse — as the post-2023 Red Sea diversions showed.

**Both sides accept:** trade and data are concentrated through a handful of chokepoints, redundancy and protection are thin, and the cost of disrupting them has fallen dramatically.

### Stress test (the benign case)

The system is more adaptable than it looks: ships rerouted around the Cape of Good Hope after 2023 (slower and costlier, but it worked), cables can be repaired and new routes built, and a Gulf ceasefire could reopen Hormuz and calm the worst of it. The vulnerability is simultaneity and speed: rerouting buys time only if one chokepoint fails at a time, and cable cuts give no warning. A coordinated or compounding event — Hormuz plus the Red Sea plus a cable cut — would overwhelm the slack the system relies on.

### Historical parallel

The 2021 grounding of a single container ship in the Suez Canal — which blocked ~12% of global trade for six days and snarled supply chains for months — showed how one chokepoint, briefly closed, ripples worldwide. The 1956 and 1967 Suez closures reordered global shipping for years. The lesson: chokepoints concentrate efficiency in good times and catastrophe in bad ones, and the world keeps rediscovering this the hard way.

### In plain terms

Almost everything you buy travels by sea, and almost all internet data travels through cables on the ocean floor — and both funnel through a few narrow passages like the Strait of Hormuz and the Red Sea. Right now several of these are in trouble at once: the Gulf conflict shut the Strait of Hormuz for months (causing an energy shock the IEA calls worse than the 1970s), the Red Sea is contested, and the undersea cables running through these same waters are being openly threatened. Ships can slowly reroute the long way around; cables can't — cutting them slows the internet, trading, and cloud services within hours. It's the most active crisis in this report, easing a little as a ceasefire reopens Hormuz, but fragile.

### What you can do now

**Expect fuel and goods prices to swing** with Gulf and Red Sea headlines; energy-shock inflation can show up at the pump and in shipping-dependent goods within weeks.

**Keep a basic resilience kit.** Because outages (including internet/communications from cable problems) are possible, the standard advice applies: some cash on hand, a little extra of essential medications and supplies, and offline copies of critical documents.

**Don't over-react in markets to each headline.** Energy and shipping shocks are real but often partly priced in; a cash buffer beats panic-trading the geopolitics.

**For businesses,** this is the case for mapping your supply chain's chokepoint exposure and holding sensible buffer inventory — just-in-time has a hidden cost when arteries clog.

**What to watch**

- The Strait of Hormuz ceasefire and reopening — and any breakdown; oil and tanker-insurance rates.
- Red Sea / Bab el-Mandeb transit volumes and any new cable-cutting incidents.
- Panama Canal water levels and the CK Hutchison port dispute; Malacca governance moves.
- Shipping rates (container and tanker) and marine insurance premiums — the price of chokepoint risk.





## Section 5 — Physical & Resource Limits

The final section grounds the whole brief in physical reality: the digital, electrified economy is colliding with hard limits in power, metal, water, and food. AI's power problem and the grid bottleneck, copper's structural deficit, the underpricing of water, and the concentrated fragility of food and fertilizer are stories where no amount of financial engineering changes the binding constraint — you cannot print electrons, copper, water, or calories.

They connect through the build-out: the same AI and electrification boom that drives the valuations two sections back and the mineral chokepoints one section back also runs straight into these physical ceilings. A data center needs power the grid cannot yet deliver; the energy transition needs more copper than is being mined; chips and agriculture need water that is increasingly scarce and mispriced; and the food system runs on a handful of fertilizer feedstocks and shipping routes. These are the slowest-moving risks in the brief and the most certain — they are arithmetic, not sentiment — which is precisely why they are under-priced. They are the long-term boundary conditions inside which every other risk plays out.

## 18. AI's power problem: the grid is the new bottleneck

**WHO THIS HITS:** Every household with an electricity bill — and every investor exposed to AI, because power, not chips, is now the limit on the whole buildout.

**CURRENT THREAT LEVEL:** **6/10 — Elevated** *The binding constraint on the AI boom*

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**Why this level:** Power has become the single biggest constraint on the AI buildout; household electricity bills are rising and turning political; and grid and transformer lead times are measured in years. **Raises it:** A heatwave-driven grid emergency tied to data-center load; sharper residential bill spikes feeding backlash and moratoria. **Lowens it:** Ratepayer-protection rules holding; on-site and behind-the-meter power scaling; a genuine demand slowdown (which early data hint at).

### What is happening

**[Established]** The constraint on artificial intelligence has shifted from chips to electricity. NVIDIA's CEO put it bluntly: every future data center will be "power-limited." US data-center electricity demand has jumped from around 23 GW in 2023 toward 42 GW in 2026, and the wait to connect a new power source to the grid has more than doubled — from about 22 months in 2000 to roughly 54 months now. Building a data center takes about two years; new transmission can take a decade. The mismatch is the story.

**[Established]** Households are paying for it. US residential electricity prices rose about 42% over five years — faster than overall inflation — and far more in data-center hotspots (Washington, D.C. up ~94%, Maryland ~74%, over five years to March 2026). In the PJM grid region, capacity prices rose a staggering ~833% between the 2024–25 and 2025–26 delivery years. The Dallas Fed estimates wholesale power prices could rise up to 50% as data-center demand doubles. It has become a political issue: at least 11 US states are weighing temporary bans on new data centers, and in March 2026 seven AI companies signed a White House-brokered "Ratepayer Protection Pledge" to fund their own grid upgrades.

**[Contested]** How binding is the limit? The IEA estimates one-fifth of global data-center investment is at risk of delay from grid bottlenecks, and Morgan Stanley sees US demand reaching 74 GW by 2028 against a ~49 GW shortfall. But Wood Mackenzie reported the buildout actually *slowing* in late 2025 (new pipeline capacity halved quarter-on-quarter), and analysts warn of "phantom" interconnection requests inflating the forecasts.

### Why it is under-covered

"AI is booming" is the headline; "the transformer has a four-year lead time" is not. The bottleneck lives in substations, interconnection queues, and utility rate cases — unglamorous, technical, and slow — even though it may decide whether the AI capex in Set 2 actually earns a return.

### The debate

**The shortage camp argues:** demand is real and exploding, the grid physically cannot keep up, and power scarcity will throttle AI, raise everyone's bills, and hand an edge to China (which already generates more than twice as much electricity as the US).

**The skeptic camp argues:** forecasts are inflated by speculative and double-counted interconnection requests; efficiency gains and a capex slowdown are already visible; and much of the "crisis" is utilities and generators talking their book into higher returns.

**Both sides accept:** grid and transformer lead times are long, residential bills are rising, and data centers are now a first-order driver of US electricity demand for the first time in decades.

### Stress test (the benign case)

If hyperscalers fund their own power (on-site gas, nuclear, behind-the-meter) and ratepayer-protection rules hold, households can be largely shielded and the buildout proceeds without a grid crisis. A demand slowdown — already hinted at — would relieve pressure further. The weakness in that case: power infrastructure is lumpy and slow, the political backlash is already here, and a single hot summer with data centers competing against air-conditioning load is the kind of event that turns a slow problem into a sudden one.

### Historical parallel

The closest analogue is the late-1990s–early-2000s telecom and dot-com infrastructure rush, when the physical buildout (fiber, then power) lagged or overshot the hype. More apt may be past episodes of demand outrunning grid investment — California’s 2000–01 electricity crisis showed how quickly a supply-demand mismatch, plus market design flaws, can produce price spikes and blackouts. The lesson: electricity is not a commodity you can summon on a tech product cycle.

### In plain terms

Building artificial intelligence takes enormous amounts of electricity, and the power grid simply can’t be expanded fast enough — new power lines and the giant transformers they need take years, while a data center goes up in about two. So power, not computer chips, is now the real limit on the AI boom. In the meantime, the extra demand is helping push up everyone’s electricity bills, especially near big data centers, which is turning into a political fight. There’s a genuine debate about whether the demand is as huge as claimed — some of it may be double-counted — and the buildout actually showed signs of slowing in late 2025.

### What you can do now

**Expect higher electricity bills** and budget for them, especially if you live near data-center growth; check whether your utility has a rate case pending.

**Look into time-of-use rates and efficiency.** Where utilities offer time-of-use pricing, shifting heavy use (laundry, EV charging) off peak can save money; basic efficiency (LEDs, insulation, a smart thermostat) is the cheapest hedge.

**If you invest in AI,** remember power is the bottleneck — the picks-and-shovels may be utilities, grid-equipment makers, and power producers as much as chipmakers; diversify rather than betting purely on the AI names.

**Engage locally if a data center is proposed near you.** Whether ratepayers or the developer pay for grid upgrades is decided in public utility-commission proceedings — those are open to comment.

### What to watch

- Grid interconnection queue times and high-voltage transformer lead times — the real speed limit.
- Residential vs data-center electricity rates, and the spread between them — who is actually paying.

- State data-center moratorium bills and ratepayer-protection rules.
- Hyperscaler capex guidance and any further signs of a buildout slowdown (the demand side of the equation).



## 19. Copper: the metal the electric future can't do without

**WHO THIS HITS:** *Anyone buying anything electric (cars, appliances, electronics), workers in construction and manufacturing, and investors — copper is in the grid, the EV, and the data center.*

**CURRENT THREAT LEVEL:** **4/10 — Moderate** *Structural deficit coming, but mostly later*

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**Why this level:** A real structural deficit is forming, but near-term the market is roughly balanced to slightly oversupplied, so the acute squeeze is later this decade. **Raises it:** Major mine disruptions stacking up (Grasberg, Kamoa-Kakula); grid + AI + defense demand outpacing supply sooner than expected. **Lowers it:** New supply and recycling ramping on schedule; weak Chinese demand persisting; substitution to aluminium where feasible.

### What is happening

**[Established]** Copper is the physical backbone of electrification — grids, motors, EVs, and data centers are all copper-hungry — and the supply side is struggling to keep up. A January 2026 S&P Global study, *Copper in the Age of AI*, projects demand rising to 42 million metric tons by 2040 (a ~50% increase) while production peaks around 2030 at ~33 million tons, leaving a ~10-million-ton gap — about 25% below demand — which it calls a “systemic risk for global industries.” Copper veteran Daniel Yergin frames the quandary neatly: copper is the great enabler of electrification, yet the pace of electrification is itself the challenge for copper.

**[Established]** Prices already reflect the tension: copper traded above \$11,700/ton (a record in December 2025) and topped \$12,000 for the first time, after a ~35% gain in 2025. The supply problem is structural — falling ore grades, rising capital costs, permitting delays, water shortages in Chile, and a thin pipeline of new mines — compounded by 2025 disruptions at giant mines (a mud intrusion shut the Grasberg mine; flooding hit Kamoa-Kakula in the DRC).

**[Contested]** The timing is genuinely debated. The ICSG sees only a small ~150,000-ton deficit in 2026, and Goldman Sachs notes a near-term surplus (with ~1 million tons sitting in US warehouses and Chinese demand soft at ~8% year-on-year in Q4 2025), forecasting the real inflection around 2029–2030. So the structural case is widely accepted; the near-term shortage is not yet here.

### Why it is under-covered

Copper is boring until it isn't. It lacks the glamour of AI or the drama of oil, and the deficit is dated to the late 2020s, so it stays a back-page commodity story — even though it is the literal wiring of every other story in this set (the grid, the data center, the EV).

### The debate

**The structural-deficit camp argues:** you cannot electrify the world without roughly doubling copper supply, and the mining industry is structurally incapable of delivering it fast enough — so scarcity and high prices are coming.

**The near-term-balanced camp argues:** right now there is no shortage — inventories are ample and Chinese demand is weak — and high prices plus recycling and substitution (aluminium in some uses) will close more of the gap than the doomers expect.

**Both sides accept:** ore grades are falling, new mines take 10–20 years to permit and build, and long-run demand from grids, AI, and defense is rising fast.

### Stress test (the benign case)

High prices are the cure for high prices: they spur exploration, recycling (which could double), substitution, and efficiency, and a prolonged Chinese slowdown (see Set 1) would soften demand. If supply responds and demand disappoints, the “deficit” arrives later and milder than feared. The vulnerability: the response time is measured in a decade-plus, ore grades keep falling regardless of price, and a cluster of mine disruptions can turn a forecast surplus into a real shortage in a single year.

### Historical parallel

The 2000s commodity supercycle is the template — Chinese industrialization drove copper from under \$2,000 to over \$10,000 a ton before supply caught up. The recurring pattern: structural demand shifts arrive faster than mines can be built, prices overshoot, capital floods in, and supply eventually responds — but the lag is long, and the interim is volatile and expensive.

#### In plain terms

Electric everything — power grids, electric cars, data centers — needs a lot of copper, and the world is on track to need far more than mines can dig up. New mines take 10 to 20 years to open, ore is getting lower quality, and big mines keep hitting problems. Experts call the looming gap a “systemic risk.” The catch: the real shortage is mostly expected later this decade (around 2029–2030); right now there’s actually plenty of copper and prices, while high, reflect the future more than the present. It’s a slow-moving squeeze, not an emergency — but it’s the wiring under almost every other story in this report.

#### What you can do now

**Mostly awareness:** expect electrification (EVs, grid upgrades, appliances) to stay expensive partly because the raw metal is getting pricier over time.

**If you invest in the energy transition or AI,** copper and other ‘picks-and-shovels’ materials are an under-appreciated way to play it — but commodities are volatile, so size positions modestly and expect a bumpy ride, not a straight line.

**Don’t chase the copper trade on a headline.** The structural story is real but the near-term market is well-supplied; buying after a price spike is how people get hurt in commodities.

**Recycle your electronics and metals.** Recycled (‘secondary’) copper is a genuine part of the solution, and e-waste recycling keeps valuable metal in the system.

### What to watch

- The copper price and exchange inventories (LME/COMEX/Shanghai) — falling stocks signal the deficit arriving.
- Major mine disruptions and production guidance (Chile, Peru, DRC, Indonesia).
- Chinese demand and stimulus — the swing factor for near-term balance.
- Mine permitting and new-project sanctioning — the only real cure for the long-run gap.





## 20. Water: the economy's most underpriced input

**WHO THIS HITS:** *Everyone who eats, and anyone near agriculture, chip plants, or data centers — water is the hidden input in food, chips, and the cloud, and it's priced as if it were free.*

**CURRENT THREAT LEVEL:** **4/10 — Moderate** *Slow and underpriced, but regionally acute*

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**Why this level:** A slow, structurally underpriced constraint that is already regionally acute (chips, farms, data centers) but lacks a single near-term systemic trigger. **Raises it:** A drought hitting a chip hub (Taiwan, Arizona) or a major breadbasket; escalation of a transboundary water dispute (Nile, Indus, Colorado). **Lowers it:** Wetter cycles easing key basins; efficiency, reuse, and — above all — sane water pricing scaling up.

### What is happening

**[Established]** Roughly half the world's population already faces water scarcity, and the Global Commission on the Economics of Water warns that, on current trends, water stress could cut global GDP by around 8% by 2050 (and far more in lower-income countries), with over half of food production at risk. Agriculture uses about 70% of the world's freshwater, so water and food are the same story. The Commission's core economic point is blunt: water is systematically *underpriced*, which encourages waste and pushes water-hungry industries into the wrong places.

**[Established]** The collision with the tech boom is direct. An analysis by the TNFD found ~40% of existing chip fabs sit in basins projected to face high or extreme water stress by 2030. Taiwan — where TSMC makes ~90% of the world's most advanced chips — suffered its worst drought since 1964 in 2020–21, cutting irrigation to ~74,000 hectares of rice so the fabs could keep running: water for chips over water for food, made explicit. Data centers add thirst on top: community opposition over water has delayed or blocked tens of billions of dollars of projects.

**[Established]** Transboundary disputes raise the stakes. The Colorado River — underpinning ~\$1.4tn of US economic activity — has run a structural deficit for over 20 years. Egypt depends on the Nile for ~97% of its water; its per-capita availability has fallen below the UN's scarcity threshold, and filling Ethiopia's Grand Renaissance Dam could sharply cut downstream supply.

### Why it is under-covered

Because water is nearly free, it doesn't show up as a market signal until it's a crisis. There is no global "water price" on a screen, so the slow erosion of this input is invisible to markets — right up until a drought stops a chip fab, empties a reservoir, or sparks a cross-border dispute.

### The debate

**The crisis camp argues:** the global water cycle is being pushed out of balance, water is dangerously mispriced, and the result is a slow-building hit to food, industry, and growth — a financial risk markets are ignoring.

**The techno-optimist camp argues:** efficiency, recycling/reuse, desalination, precision irrigation, and (eventually) better pricing can stretch supply enormously, and the largest firms can secure their own water — so it's a manageable engineering and policy problem, not a catastrophe.

**Both sides accept:** water is underpriced, agriculture dominates use, and water-intensive industries are increasingly sited in already-stressed basins.

### Stress test (the benign case)

Water problems are mostly local and solvable: efficiency and reuse have huge headroom, chipmakers already recycle most process water, and pricing reform (where politically possible) would reallocate water quickly. Most of the world is not at the extreme tail. The vulnerability is that water is intensely political and local: you cannot ship it far or cheaply, drought is getting more frequent, and a single dry year in the wrong basin — Taiwan, Arizona, the Nile — can disrupt chips, food, or geopolitics with little warning.

### Historical parallel

History is full of water-driven economic and political upheaval, from the US Dust Bowl of the 1930s to Cape Town's 2018 "Day Zero" near-miss and Chennai running dry in 2019. The recurring lesson: societies treat water as infinite until a sharp shortage forces brutal rationing and reveals how much of the economy was silently resting on it.

#### In plain terms

Water is the invisible ingredient in food, computer chips, and data centers — and because it's priced as though it were free, we waste it and build thirsty industries in dry places. Farming uses about 70% of fresh water, so water shortages are really food shortages. The tech boom collides with this directly: Taiwan, which makes most advanced chips, once cut water to rice farms during a drought so the chip factories could keep running. None of this is a single global emergency — it's slow and very local — but a bad drought in the wrong place (a chip hub, a major farming region, a shared river) can cause outsized economic damage quickly.

#### What you can do now

**This is mostly awareness** — but understanding that food prices, chip supply, and some regional economies hinge on water helps you read the news (a drought story can be an economic story).

**If you live in a water-stressed region**, water-efficient appliances, fixing leaks, and drought-tolerant landscaping cut bills and exposure — and water rates are likely to rise over time as scarcity gets priced in.

**Factor water into big location decisions.** If you're buying property or a business, a region's long-term water situation is a real and under-considered risk.

**For investors**, water (utilities, efficiency, treatment) is a long-duration theme, and water stress is a genuine risk hidden inside agriculture, semiconductors, and data-center holdings — worth checking, not panicking over.

### What to watch

- Drought conditions in key chip and farming regions (Taiwan, US Southwest, India, Brazil).
- Reservoir levels on stressed systems (Colorado River/Lake Mead) and major dam disputes (Nile/GERD, Indus).
- Moves toward real water pricing, water markets, and reuse mandates — the structural fix.
- Data-center and fab approvals being blocked or relocated over water — the economic bite.



## 21. Food and fertilizer: a concentrated, fragile system

**WHO THIS HITS:** *Everyone who eats — especially lower-income households and import-dependent countries — and farmers facing record input costs. Fertilizer shocks become food-price shocks within a year.*

CURRENT THREAT LEVEL: **6/10 — Elevated** *Active export curbs and Gulf-input risk*

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**Why this level:** Active export curbs (China through August, Russia controls), Strait of Hormuz disruption to fertilizer inputs, and an FAO warning that staple-crop output could fall within 6–9 months keep this elevated. **Raises it:** A prolonged Hormuz/Red Sea disruption to sulfur and ammonia; a poor harvest; new export bans cascading across producers. **Lowers it:** Export curbs lifting on schedule; new fertilizer capacity arriving in 2026–27; a calm Gulf and a good growing season.

### What is happening

**[Established]** The world's food supply rests on a handful of fertilizer producers and trade routes, and several are constrained at once. China — a top phosphate and nitrogen exporter — has restricted fertilizer exports (quotas and licensing) into late August 2026 to protect domestic supply (its 2024 nitrogen exports fell more than 90% year-on-year). Russia, a dominant nitrogen exporter, suspended export licenses for ammonium nitrate; Belarusian potash remains under sanctions; and the EU has tariffed Russian and Belarusian fertilizer. Prices sit roughly 17–20% above year-ago levels.

**[Established]** The 2026 Gulf conflict makes it worse. Large volumes of urea, ammonia, phosphates, and sulfur — critical fertilizer inputs — move through the Strait of Hormuz, and the FAO warns that reduced availability could lower output of wheat, maize, and rice within 6–9 months and push food prices up. Countries are scrambling for workarounds: India locked in Canadian and rerouted Russian supply; the US expanded sanctions waivers for Venezuelan fertilizer.

**[Established]** Underneath the headlines sit two structural concentrations. Canada supplies roughly 90% of US potash imports — one analyst's blunt summary: the US “cannot feed itself... without either Canada or Russia.” And phosphate is doubly squeezed: China is diverting it into lithium-iron-phosphate EV batteries (a direct link to electrification), while Morocco holds the lion's share of the world's phosphate-rock reserves. **[Contested]** Estimates that global phosphorus production could peak around 2035 are debated, but the concentration of reserves is not.

### Why it is under-covered

Fertilizer is invisible until food gets expensive. It's a chemistry-and-logistics story most people never think about — yet roughly half the world's food production depends on synthetic fertilizer, and a fertilizer shock today is a grocery-bill and hunger shock 6–12 months later, especially in poorer importing countries (a direct link to the Set 1 developing-debt story).

### The debate

**The fragility camp argues:** production and trade are dangerously concentrated in a few countries and chokepoints, export bans are spreading, and the Gulf conflict plus phosphate scarcity set up a multi-year food-security risk.

**The reassurance camp argues:** fertilizer markets are adapting (rerouting, new capacity coming in 2026–27), affordability relative to food prices is still near pre-2020 norms, and bans tend to be temporary — so this is volatility, not collapse.

**Both sides accept:** supply is concentrated, prices are elevated, key inputs run through contested waters, and the lag from fertilizer to food means today's disruptions show up in next year's harvests.

### Stress test (the benign case)

The system has proven adaptable: after the 2022 shock, trade rerouted and prices eventually fell. New capacity (potash in Canada, phosphate elsewhere) is arriving, export bans are usually lifted within a season, and high prices pull supply forward. The vulnerability is correlation and timing: if a prolonged Gulf disruption hits sulfur/ammonia during planting season while China's curbs persist and a major harvest disappoints, the buffers thin fast — and the people hurt first are those who can least afford it.

### Historical parallel

2022 is the fresh memory: Russia's invasion of Ukraine spiked fertilizer and grain prices, drove food inflation worldwide, and contributed to unrest in import-dependent countries. Further back, the 2007–08 food-price crisis — amplified by export bans — triggered riots across dozens of countries. The pattern: fertilizer and grain shocks transmit into political instability with a lag, and export bans make a shortage worse by turning it into a panic.

### In plain terms

Most of the world's food depends on fertilizer, and fertilizer comes from just a few countries and travels through a few sea routes — several of which are now restricted or contested at once. China is limiting exports into late summer, Russia has curbed some too, and key fertilizer ingredients pass through the Strait of Hormuz, which the Gulf conflict has disrupted. The UN warns this could lower wheat, corn, and rice harvests within 6–9 months and raise food prices. There's also a deeper vulnerability: the US gets about 90% of one key fertilizer (potash) from Canada, and the world's phosphate is heavily concentrated in a few places — with China now diverting some into EV batteries. It tends to be volatility rather than collapse, but it hits grocery bills, and the poorest countries first.

### What you can do now

**Expect food-price volatility** over the next year and build a little flexibility into your grocery budget; staples (grains, cooking oil) are the most exposed.

**A modest food buffer is reasonable, panic-buying is not.** A couple of weeks of shelf-stable staples is sensible household resilience; hoarding makes shortages worse for everyone (that's exactly how the 2008 crisis spiraled).

**Reduce food waste.** Roughly a third of food is wasted; cutting your own waste is the most direct way to blunt higher prices and ease pressure on the system.

**If you garden or can buy local,** even a small amount of home or local food production adds personal resilience to global supply shocks.

### What to watch

- Fertilizer prices (urea, DAP, potash) and the FAO Food Price Index — the leading and lagging gauges.
- China's export-curb expiry (late August 2026) and any Russian/other new restrictions.

- Strait of Hormuz / Red Sea status as it affects sulfur and ammonia flows.
- Harvest and planting reports in major exporters (US, Brazil, Black Sea, India) — where shortages become real.





## Pattern analysis: how the five mechanisms interlock

Read together, the twenty-one stories are not twenty-one separate problems but five layers of one system — and grouping them by mechanism is what makes the layers, and the wiring between them, visible. The simplest way to see the wiring is to follow a shock. A spark — most plausibly from the geoeconomic chokepoints or the credit and valuation gaps — raises volatility. That volatility is the precise trigger for the leverage layer: higher margins and haircuts force the basis trade, the carry trade and the clearinghouses to sell into a falling market. The forced selling drains funding and feeds back into credit. And the public backstops that stopped this loop in 2008 and 2020 are now constrained by weakened sovereign balance sheets. Underneath all of it, physical limits are the slow boundary that makes several of these shocks more likely over time.

The tightest coupling is between Funding & Leverage and Credit, Consumers & Valuations — two halves of the same reflexive loop. Leverage assumes the assets it is borrowed against are worth their marks, and those marks assume funding stays cheap. When a credit event forces a mark lower — an AI-financing wobble, a failed commercial-real-estate refinancing — it delivers exactly the volatility spike that detonates the leverage; and when leverage unwinds, the fire-sale prices it produces become the new, lower marks. Each section is the other's trigger.

Geoeconomics and the sovereign-monetary layer set the conditions around that loop. Geoeconomics supplies the most likely exogenous spark — a closed strait, a minerals embargo, an export ban on the machines that make advanced chips — because these are deliberate acts of state, not market accidents, and they arrive on political rather than financial timetables. Sovereigns and the monetary order, meanwhile, determine how much firefighting capacity remains: every unit of fiscal space lost to debt, demographics, or a creeping loss of faith in the dollar is a unit not available to backstop the next funding run. The same gold-buying and de-dollarization that hedge that loss of faith also signal it.

Physical & Resource Limits is the floor the others stand on. The AI and electrification boom that powers the valuations and the mineral chokepoints runs straight into hard limits on power, copper, water and food — limits that are arithmetic, not sentiment, and therefore both the most certain and the most under-priced risks here. The honest synthesis is this: very few of these twenty-one stories are flashing red on their own today. The systemic risk lives in the correlations — that a single shock can light several layers at once, and that the firefighters now carry less water than they did. That is precisely why the companion tracker scores these stories by mechanism and watches the cross-section aggregates: the number that matters most is not any single story, but how many layers move together.

## Master watchlist

The highest-signal indicators to watch, by mechanism. Most are public and free; the live tracker monitors many of them automatically.

### Funding & Leverage

- The MOVE index: sustained spikes in Treasury volatility are what trigger the margin-and-haircut doom loops in the basis and carry trades.
- Repo rates and the SOFR-fed funds spread at quarter-ends: early signs of dealer balance-sheet strain and tightening funding.
- SEC Treasury central-clearing milestones (2026–2027) and Bank of Japan policy with 30-year JGB yields: structural shifts in the leverage backdrop.
- Large stablecoin redemptions or a depeg: the crypto-native version of a funding run.

### Credit, Consumers & Valuations

- AI vendor-financing and special-purpose-vehicle disclosures: the circularity becomes visible when revenue assumptions slip.
- Commercial-real-estate refinancing volumes and regional-bank CRE exposure: where the maturity wall meets thin capital.
- Subprime-auto and buy-now-pay-later delinquencies: the consumer stress that does not show in headline credit data.
- Insurer non-renewals and withdrawals in exposed states: the quiet erosion of home insurability and value.

### Sovereigns & the Monetary Order

- Emerging-market sovereign spreads and new IMF programs: where developing-world debt distress surfaces first.
- The French OAT–Bund spread and budget votes: the eurozone's nearest fiscal-political fault line.
- Official gold purchases and reserve-currency shares: the pace of the hedge against the dollar.
- Pension funding ratios and forced contribution increases: the slow, large strain on public balance sheets.

### Geoeconomics & Trade Weapons

- Rare-earth and critical-mineral export curbs or quotas: availability is not access.
- ASML order flow and chip export-control announcements: the lithography chokepoint in motion.
- War-risk and shipping insurance rates through Hormuz, the Red Sea and the Taiwan Strait: the price of sea-lane risk.
- China's producer-price index and export prices: the gauge of exported deflation.

### Physical & Resource Limits

- Data-center power-queue and grid-interconnection backlogs: the binding limit on the AI build-out.
- Copper inventories and mine-supply guidance: the structural deficit behind electrification.
- Reservoir and aquifer levels in chip-fabrication and farming regions: the underpriced water constraint.

- Fertilizer feedstock (natural gas, ammonia) prices and grain-export concentration: the fragility in the food system.

## Confidence, caveats and what could make this wrong

A word on confidence. Throughout this brief, claims are tagged [Established] (documented by multiple authoritative sources), [Contested] (real but debated in size or interpretation), [Single-source] (reported but not yet widely corroborated), or [Forecast] (a forward projection, not a fact). The threat levels are analytical judgments on a 1–10 scale, not probabilities, and they describe vulnerability — how much damage a position could do if stressed — not a prediction that it will be. Several of the highest-scored stories may sit quietly for years.

This brief is deliberately organized by mechanism rather than by headline, and that lens is itself a claim: that these twenty-one risks are best understood as five layers of one system whose real danger lies in their correlations. That framing could be wrong in either direction — a shock could stay contained in one layer, or a connection we have not drawn could matter more than the ones we have. The material is informational analysis, not investment, legal or financial advice; it points to what to watch and why things relate, and leaves the decisions to you.

## Plain-language glossary

**Basis trade** — a leveraged trade that profits from the small price gap between US Treasury bonds and Treasury futures, financed with very short-term borrowing.

**Repo** — the overnight market where institutions borrow cash against securities; the funding that keeps much of the leverage in this brief alive.

**Haircut** — the safety margin a lender demands on collateral. A zero haircut means no cushion if prices move against the borrower.

**MOVE index** — a gauge of expected US Treasury-market volatility — the bond market's equivalent of the VIX.

**Carry trade** — borrowing in a cheap, low-rate currency (often the yen) to invest in higher-yielding assets elsewhere; profitable until the funding currency moves.

**Private credit** — loans made by non-bank funds directly to companies, outside public markets and with less frequent valuation.

**Stablecoin** — a crypto token pegged to the dollar, typically backed by reserves that increasingly include US Treasuries.

**Clearinghouse (CCP)** — an institution that stands between buyers and sellers to guarantee trades; efficient in normal times, a concentrated single point of failure in a crisis.

**Maturity wall** — a cluster of debt all coming due in a short window, forcing refinancing at whatever rates then prevail.

**Mark-to-market** — valuing an asset at its current market price rather than its original cost; 'extend and pretend' delays this reckoning.

**Credit spread (OAS)** — the extra yield investors demand to hold risky debt over safe government debt; widening spreads signal rising default fear.

**Sovereign spread** — the gap between one government's borrowing cost and a benchmark (e.g., French OATs over German Bunds); a market gauge of fiscal risk.

**De-dollarization** — the gradual shift by central banks and governments away from the US dollar as a reserve and settlement currency, often toward gold.

**Chokepoint** — a narrow node — a strait, a mine, a single supplier — through which a large share of global supply must pass, giving whoever controls it leverage.

**Grid interconnection** — the queue and physical process for connecting new power demand or supply to the electricity grid; a binding bottleneck for AI data centers.

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**End of Set 1.** If you judge these seven to be heavy enough in weight, the next set can lead with insurance-market retreat as a systemic risk, China's local-government debt machinery in depth, and the eurozone's fiscal-political strains — plus any thread above you want taken further.

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**End of Set 2.** These seven were chosen to reach everyday life — your savings, home, car, and retirement. If you judge them heavy enough, the next set can go to global food and commodity stress (cocoa, coffee, copper for electrification), the power-grid bottleneck behind the AI boom, water scarcity as an economic force, or any thread above taken deeper. And if you'd like, I can retrofit Set 1 with these same plain-language and action sections.

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**End of Set 3.** This set was the physical foundation under Sets 1 and 2. Per our earlier discussion, the clearly-strong material is starting to thin — the next set would likely cover money-market funds and the repo system, the offshore-dollar (eurodollar) market, semiconductor materials beyond lithography, and demographic collapse in specific countries (Korea, Italy) — strong topics, but several rhyme with mechanisms we've already covered. A good moment to consider shifting to periodic updates (via the tracker) plus a smaller 'what's newly emerged' set, rather than chasing volume. Your call.